UNIVERSITÀ DEGLI STUDI DI VERONA

DEPARTMENT OF ECONOMICS

Master's Degree in Economics and Data Analysis

DO MACHINE LEARNING MODELS IMPROVE STOCK RETURN PREDICTION?

Evidence from S&P 500 Constituent Markets and Dimension Sensitivity (2015–2024)

Supervisor:

Prof. Giuseppina Chesini

Candidate:

Murtuza Yusuf Rangwala

Matricola: VR508566

Academic Year 2024/2025

Abstract

Does applying modern machine learning and deep sequence models to stock market data yield genuine predictive advantage over traditional linear asset pricing benchmarks? To answer this question, I evaluate a broad range of quantitative models using a 10-year point-in-time panel of S&P 500 constituent equities from 2015 through 2024 (comprising 2,516 daily cross-sections). The predictive feature set includes 59 variables spanning technical signals, quarterly accounting ratios, macroeconomic state variables, market sentiment indicators, and rolling factor risk exposures.

To ensure realistic empirical evaluation without lookahead contamination, I implement a 5-fold expanding-window walk-forward validation framework across the 2020–2024 out-of-sample period. Empirical results demonstrate that deep sequence models outperform classical linear estimators (Fama-MacBeth OLS, regularized LASSO) as well as static decision tree ensembles (Random Forest, XGBoost). Among all tested architectures, the custom **Temporal Fusion Deep Multimodal Gated Attention (TFD-MGA)** model achieves the strongest predictive performance, recording an out-of-sample daily Information Coefficient (IC) of $+0.0348$, an Information Ratio ($ICIR$) of 3.12, a ROC AUC of 0.6120, and a directional accuracy of 56.84%. A formal Diebold-Mariano test confirms that these gains over baseline PyTorch LSTM models are statistically significant ($DM = 2.41, p = 0.016$).

When evaluated in simulated daily trading portfolios accounting for 10 bps round-trip transaction costs and 1-day execution delays, an initial $1,000 USD deposit starting in January 2020 grows to $3,120.99 USD under a baseline LSTM long strategy. Introducing a 2:1 Take-Profit/Stop-Loss risk overlay enhances capital accumulation to $4,368.50 USD for LSTM and $6,482.10 USD for TFDMGA. However, Fama-French 5-factor spanning regressions reveal an estimated net strategy alpha of $\hat{\alpha} = -0.18\%$ per year ($p = 0.976, R^2 = 41.2\%$). Because net alpha is statistically zero, the returns produced by these deep learning models represent dynamic compensation for systematic factor risk exposures rather than unpriced market inefficiency.

Keywords: Machine Learning, Empirical Asset Pricing, Cross-Sectional Return Prediction, Deep Learning, Transformer Attention, Fama-French Spanning, Transaction Costs, Walk-Forward Optimization.

يا رب زدني علما

“And say, ‘My Lord, increase me in knowledge.’ ”

, The Holy Qur’an, Surah Ta-Ha (20:114)

Acknowledgements

I wish to express my deepest appreciation to my supervisor, **Prof. Giuseppina Chesini**, whose thoughtful mentorship, advice, and academic support guided me throughout the preparation of this Master's thesis. Her insights helped shape the research direction and refine the empirical methodology presented in these pages.

My sincere thanks extend to the professors and academic staff of the **Department of Economics at the University of Verona**. The knowledge, rigorous training, and encouragement I received during my Master's studies played a vital role in my academic development.

Studying at the **University of Verona** has been a transformative experience. Moving from Mumbai, India, to Italy opened doors to a vibrant international community and enriched my perspective both academically and personally. I am immensely grateful to the university for providing this platform.

I am equally indebted to **ESU Verona and the Borsa di Studio scholarship programme**. Their financial assistance granted me the security needed to concentrate fully on my studies and complete my degree without financial burden.

To my classmates and close friends in Italy, thank you for making Verona feel like a second home. The shared study sessions, discussions, and personal support through every challenge created memories that I will treasure for years to come.

I also recognize the broader open-source research community, whose computational tools, software libraries, and open frameworks enabled the empirical implementation of this work.

Above all, I dedicate this work to my family. I am profoundly thankful to my father, whose belief in my potential inspired me to pursue higher education abroad. His sacrifices, advice, and unconditional support gave me confidence during difficult moments. My heartfelt thanks also go to my mother, my sister, and my extended family for their love, patience, and constant encouragement throughout this entire journey.

Murtuza Rangwala
Master of Economics and Data Analysis
University of Verona

Contents

List of Figures

List of Tables

1. Introduction

1.1 Background and Economic Motivation

Why do financial asset returns vary across stocks, and how effectively do market prices incorporate available information? Answering this fundamental question lies at the heart of empirical asset pricing (Fama, 1970; Fama and French, 1993). For decades, academic researchers modeled expected equity returns through linear factor frameworks, evolving from Markowitz's (1952) mean-variance framework to the Capital Asset Pricing Model (CAPM) of Sharpe (1964) and Lintner (1965), Ross's (1976) Arbitrage Pricing Theory, Carhart's (1997) four-factor momentum specification, Pástor and Stambaugh's (2003) liquidity model, Hou, Xue, and Zhang's (2015) q-factor framework, and Fama and French's (2015) five-factor model. These classic models share a core assumption: stock returns are governed linearly by a small set of static risk exposures.

Real-world financial markets, however, display non-linearities, heavy-tailed return distributions, and dynamic regime shifts. Firm-level characteristics interact in nonlinear ways, for instance, valuation metrics behave differently in market drawdowns compared to economic expansions (Gu et al., 2020; Kozak et al., 2020). Standard Ordinary Least Squares (OLS) regressions face structural hurdles in high dimensional panels, risking omitted variable bias when specifications are too simple, or variance inflation when dozens of firm characteristics are included simultaneously.

Machine learning (ML) and deep sequence architectures provide a flexible framework for modeling nonlinear characteristic interactions and temporal patterns in equity markets (Gu et al., 2020, 2021; Chen et al., 2024; Kozak et al., 2020; Freyberger et al., 2020; Kelly et al., 2019). Techniques ranging from regularized linear models (Tibshirani, 1996; Zou and Hastie, 2005) and decision tree ensembles (Breiman, 2001; Friedman, 2001; Chen and Guestrin, 2016) to temporal convolutional networks (Bai et al., 2018) and multimodal Transformer architectures (Vaswani et al., 2017; Lim et al., 2021) allow researchers to extract predictive signals directly from raw panel data.

Yet applying machine learning to financial asset pricing differs from standard computer vision or language modeling. Financial return series are characterized by low signal-to-noise ratios, regime shifts, and structural breaks (López de Prado, 2018, 2020). Naive applications risk lookahead bias, survivorship bias, multiple testing inflation (Harvey et al., 2016; Harvey and Liu, 2020; Hou et al., 2020), and unmodeled transaction cost drag (Novy-Marx and Velikov, 2016; McLean and Pontiff, 2016; Avramov et al., 2023). Without out-of-sample walk-forward validation and trading friction modeling, apparent predictive gains can quickly disappear in live trading.

In this thesis, I evaluate machine learning models for cross-sectional return prediction across S&P 500 equities over the 2015–2024 period, addressing these methodological challenges directly.

1.2 Research Questions and Core Hypotheses

To evaluate the statistical predictability and economic viability of machine learning models, I frame my investigation around four central research questions:

Research Question 1 (Statistical Predictability): Do nonlinear machine learning models (Random Forest, XGBoost, LSTM, TFDMGA) achieve statistically significant out-of-sample return predictability compared to linear econometric baselines (Fama-MacBeth OLS, LASSO, ElasticNet)?

> *Hypothesis 1*: Nonlinear models, especially sequence-aware deep neural networks (Bai et al., 2018; Chen et al., 2024), outperform linear baselines in out-of-sample Information Coefficient (IC), Information Ratio ($ICIR$), and Directional Accuracy by capturing nonparametric characteristic interactions and temporal dependencies.

Research Question 2 (Multi-Modal Information Integration): Does integrating technical indicators, quarterly Bloomberg accounting ratios, macroeconomic state variables, and news/social sentiment within a dynamic gating architecture improve prediction accuracy over single-modality models?

Hypothesis 2: Multimodal feature integration (Cong et al., 2021; Lim et al., 2021) allows dynamic attention mechanisms to reweight feature modalities as macroeconomic regimes shift, yielding higher out-of-sample correlation and signal stability.

Research Question 3 (Economic Significance & Transaction Friction): Can out-of-sample return forecasts generate profitable trading strategies after accounting for transaction frictions (10 bps round-trip fees, execution delays, turnover, and stop-loss risk management)?

Hypothesis 3: While daily portfolio rebalancing incurs cost drag under 10 bps transaction fees (Novy-Marx and Velikov, 2016; Avramov et al., 2023), deep sequence models paired with dynamic take-profit/stop-loss overlays help control drawdowns and compound capital over time.

Research Question 4 (Econometric Factor Spanning & Market Efficiency): Are out-of-sample returns produced by deep learning models indicative of unpriced market alpha ($\alpha > 0$), or are they fully spanned by systematic risk factors ($\alpha = 0$) within the Fama-French 5-Factor framework?

Hypothesis 4: Out-of-sample strategy returns are explained by systematic factor risk exposures ($Mkt - RF, SMB, HML, RMW, CMA$) (Fama and French, 2015; Kelly et al., 2019), indicating that machine learning models act as dynamic factor allocation engines rather than discoverers of unpriced arbitrage.

1.3 Summary of Key Empirical Findings

Using a 5-fold expanding-window walk-forward validation scheme over the 2020–2024 out-of-sample period (detailed in Chapter 3 and Chapter 4), my main findings are:

1. **Performance in Return Rank Forecasting**: The custom Temporal Fusion Deep Multimodal Gated Attention (**TFDMGA**) network achieves the highest predictive

accuracy among all evaluated models, recording an out-of-sample daily Information Coefficient of $IC = +0.0348$, an Information Ratio of $ICIR = 3.12$, a ROC AUC of 0.6120, and a directional accuracy of 56.84%.

2. **Statistical Comparison with Baseline Sequence Models**: A formal Diebold and Mariano (1995) forecast accuracy test with Harvey et al. (1997) small-sample corrections confirms that TFDMGA yields statistically significant performance gains over a baseline PyTorch LSTM model ($DM = 2.41, p = 0.016$), validating the structural benefits of crossmodal ring attention and dynamic macro gating.

3. **Capital Compounding Trajectory**: Under a realistic 10 bps round-trip transaction cost setting and a 1-day execution buffer (Novy-Marx and Velikov, 2016), an initial $1,000 USD deposit in January 2020 grows to **$3,120.99 USD** under the baseline LSTM Q_5 long strategy, and reaches **$4,368.50 USD** with a 2:1 Take-Profit/Stop-Loss (TPSL) risk overlay. Under TFDMGA with the TPSL overlay, capital compounds to **$6,482.10 USD** by December 2024 (see Table 4.5 in Chapter 4).

4. **Fama-French 5-Factor Spanning & Factor Timing**: Spanning regressions against the Fama and French (2015) 5-factor model yield an estimated net strategy alpha of $\hat{\alpha} = -0.18\%$ per year ($p = 0.976, R^2 = 41.2\%$). Because net alpha is statistically zero, strategy returns are explained by dynamic risk exposure to systematic factor risks, specifically operating profitability (RMW beta $= +0.512, t = +9.12$), consistent with market efficiency (Fama, 1970).

1.4 Structure of the Thesis Manuscript

The remainder of this thesis is structured across four consolidated chapters:

- **Chapter 2 (Literature Review)** surveys academic literature in empirical asset pricing, factor models, high dimensional regularization, and financial machine learning.

- **Chapter 3 (Data & Methodology)** details the Bloomberg point-in-time data pipeline, constituent panel construction, 59-variable feature taxonomy, baseline model formulations, and the custom TFDMGA network architecture.
- **Chapter 4 (Empirical Results & Backtesting)** presents daily Fama-MacBeth baselines, LASSO shrinkage derivations, SHAP interpretability analysis, out-of-sample predictive metrics, ablation results, trading backtests under transaction fees, and Fama-French spanning regressions.
- **Chapter 5 (Conclusion & Future Research Extensions)** summarizes the main findings, discusses theoretical and practical implications, addresses study limitations, and outlines future research extensions.

2. Literature Review

2.1 Introduction

Understanding stock return predictability requires tracing the evolution of empirical asset pricing over the past six decades. Early academic frameworks established that asset prices in competitive markets reflect available information (Fama, 1970), modeling expected excess returns as linear functions of systematic risk exposures (Sharpe, 1964; Lintner, 1965). Subsequent econometric advances operationalized these principles through multifactor cross-sectional regression models (Fama and MacBeth, 1973; Ross, 1976; Fama and French, 1993, 2015).

However, the rapid expansion of financial data led to what Cochrane (2011) famously termed the "factor zoo", a proliferation of over 300 candidate anomaly variables claiming to generate alpha. In response, modern financial econometrics has developed along two primary paths: (i) statistical auditing of data-mining inflation and false discovery rates (Harvey et al., 2016; Harvey and Liu, 2020; Hou et al., 2020), and (ii) high dimensional regularized estimation and nonparametric machine learning (Kozak et al., 2020; Freyberger et al., 2020; Gu et al., 2020; Chen et al., 2024).

In this chapter, I synthesize the asset pricing literature around six key thematic pillars that ground the theoretical framework, model architectures, feature selection, and statistical inference used in this thesis:

1. **Theoretical Asset Pricing Foundations**: From the single factor equilibrium of CAPM and APT to the multifactor models of Fama-French, Carhart, and Pastor-Stambaugh.

2. **The Factor Zoo, Multiple Testing, and Shrinkage**: Multiple testing inflation, false discovery rates, and high dimensional regularized SDF estimation.

3. **High-Dimensional SDF Estimation & Characteristic Interactions**: Nonparametric characteristic selection, latent factor models, and dynamic risk loadings.

4. **Machine Learning & Deep Learning Paradigms**: Nonlinear empirical asset pricing, instrumented principal components, and deep sequence SDF models.

5. **Financial Time-Series Econometrics & Inference**: HAC covariance estimation, out-of-sample forecast accuracy tests, and data-mining diagnostics.

6. **Financial ML Protocols & Overfitting Mitigation**: Backtesting rules, expanding walk-forward splits, microstructural frictions, and transaction cost frameworks.

2.2 Theoretical Asset Pricing Foundations

2.2.1 Equilibrium Foundations: The CAPM and Arbitrage Pricing Theory

Modern quantitative finance originates with the mean-variance portfolio theory of Markowitz (1952) and the Capital Asset Pricing Model (CAPM) developed by Sharpe (1964) and Lintner (1965). Under the assumptions of homogeneous investor expectations, frictionless markets, and quadratic utility or normal return distributions, investors hold identical portfolios of risky assets weighted by market capitalizations. As a result, the market portfolio R_m is mean-variance efficient, and an asset's expected return is linearly related to its covariance with the aggregate market portfolio:

$$\mathbb{E}[R_{i,t}] - R_{f,t} = \beta_{i,m} \left(\mathbb{E}[R_{m,t}] - R_{f,t}\right) \tag{2.1}$$

where the market beta $\beta_{i,m} = \frac{\mathrm{Cov}(R_{i,t}, R_{m,t})}{\mathrm{Var}(R_{m,t})}$ measures systemic market risk. Black (1972) relaxed the assumption of risk-free borrowing, introducing the zero-beta CAPM and showing that the security market line remains linear but exhibits a higher intercept and flatter slope when borrowing constraints bind.

Recognizing the strict equilibrium assumptions of the CAPM, Ross (1976) formulated the Arbitrage Pricing Theory (APT). Based on the absence of asymptotic arbitrage rather than general equilibrium, APT posits that asset returns follow a K-factor linear generative structure:

$$R_{i,t} = \mathbb{E}[R_{i,t}] + \sum_{k=1}^{K} \beta_{i,k} f_{k,t} + \epsilon_{i,t} \tag{2.2}$$

where $f_{k,t}$ represent mean-zero common macroeconomic risk factors, $\beta_{i,k}$ are factor loadings, and $\epsilon_{i,t}$ is an idiosyncratic error satisfying $\mathrm{Cov}(\epsilon_{i,t},\, \epsilon_{j,t}) = 0$ for all $i \neq j$. By applying linear pricing space arguments in infinite-dimensional Hilbert spaces, Ross showed that expected returns are linearly spanned by factor risk premia λ_k:

$$\mathbb{E}[R_{i,t}] - R_{f,t} = \sum_{k=1}^{K} \beta_{i,k}\lambda_k \tag{2.3}$$

2.2.2 The Fama-MacBeth Two-Pass Regression Framework

To empirically test whether factor loadings explain the cross-section of expected returns, Fama and MacBeth (1973) established the classic two-pass regression methodology. In the first pass, time series regressions are conducted for each asset i over an estimation window T_1 to estimate factor loadings $\hat{\beta}_{i,k}$:

$$R_{i,t} - R_{f,t} = \alpha_i + \sum_{k=1}^{K} \beta_{i,k} f_{k,t} + \epsilon_{i,t}, \quad t = 1, \ldots, T_1 \tag{2.4}$$

In the second pass, cross-sectional regressions are executed at each time step t over an evaluation window T_2 using the estimated factor loadings as explanatory variables:

$$R_{i,t} - R_{f,t} = \gamma_{0,t} + \sum_{k=1}^{K} \hat{\beta}_{i,k}\gamma_{k,t} + u_{i,t}, \quad i = 1, \ldots, N \tag{2.5}$$

The point estimates of factor risk premia $\bar{\gamma}_k$ and their standard errors are obtained by averaging the cross-sectional slope coefficients across time:

$$\bar{\gamma}_k = \frac{1}{T_2} \sum_{t=1}^{T_2} \hat{\gamma}_{k,t}, \quad \mathrm{SE}(\bar{\gamma}_k) = \sqrt{\frac{1}{T_2(T_2 - 1)} \sum_{t=1}^{T_2} (\hat{\gamma}_{k,t} - \bar{\gamma}_k)^2} \tag{2.6}$$

While standard Fama-MacBeth standard errors assume cross-sectional independence and zero serial correlation, modern implementations incorporate Heteroskedasticity and Autocorrelation Consistent (HAC) corrections (Newey and West, 1987) and Shanken (1992) corrections for generated regressor error in $\hat{\beta}_{i,k}$.

2.2.3 Multi-Factor Models: Fama-French, Carhart, and Pastor-Stambaugh

Empirical challenges to the single factor CAPM increased during the late 1980s and early 1990s as cross-sectional anomalies emerged. Fama and French (1992) showed that firm size (market capitalization) and book-to-market equity ratio (B/M) absorbed much of the cross-sectional explanatory power of market beta. To formalize these empirical findings, Fama and French (1993) introduced the Fama-French Three-Factor Model:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}(R_{m,t} - R_{f,t}) + \beta_{i,2}SMB_t + \beta_{i,3}HML_t + e_{i,t} \tag{2.7}$$

where SMB_t (Small Minus Big) captures the size effect, and HML_t (High Minus Low) captures the value effect.

Observing that the three-factor model failed to price the cross-sectional price momentum phenomenon identified by Jegadeesh and Titman (1993), Carhart (1997) expanded the framework into a Four-Factor Model by adding a momentum factor (WML_t, Winners Minus Losers):

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}(R_{m,t} - R_{f,t}) + \beta_{i,2}SMB_t + \beta_{i,3}HML_t + \beta_{i,4}WML_t + e_{i,t} \tag{2.8}$$

In parallel, Pástor and Stambaugh (2003) showed that market-wide liquidity is an undiversifiable risk factor. Stocks with high sensitivity to aggregate market liquidity fluctuations command higher expected returns to compensate investors for illiquidity shocks during market downturns.

To address profitability and investment anomalies, Fama and French (2015) extended their three-factor model to a Five-Factor Asset Pricing Model:

$$\begin{aligned} R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}(R_{m,t} - R_{f,t}) + \beta_{i,2}\, SMB_t + \beta_{i,3}\, HML_t \\ + \beta_{i,4}\, RMW_t + \beta_{i,5}\, CMA_t + e_{i,t} \end{aligned} \tag{2.9}$$

where RMW_t (Robust Minus Weak) measures the profit margin return differential, and CMA_t (Conservative Minus Aggressive) captures investment intensity differentials. Grounded theoretically in the dividend discount model, Fama and French (2015) showed that holding book-to-market fixed, higher expected earnings imply higher expected returns, while higher expected investment rates lower expected returns. Concurrently, Hou et al. (2015) established the q-factor model ($Mkt, ME, I/A, ROE$), showing that investment and profitability factors derived from q-theory span a wide range of cross-sectional anomalies.

2.3 The Factor Zoo Controversy, Multiple Testing, and Shrinkage

2.3.1 The Proliferation of Anomalies and Cochrane's "Factor Zoo"

By the early 2010s, academic literature had documented over 300 firm characteristics claiming to predict stock returns. In his American Finance Association presidential address, Cochrane (2011) famously diagnosed this proliferation as the "factor zoo." Cochrane noted that while early theoretical models assumed a small number of macro risk factors span expected returns, the literature had created a large collection of firm characteristics without establishing whether candidate factors were incremental to existing benchmarks or merely duplicate representations of underlying noise.

This issue was compounded by post-publication return decay. McLean and Pontiff (2016) examined 97 published cross-sectional anomalies and documented that out-of-sample anomaly portfolio returns decline by an average of 26% after academic publication and up to 58% after original sample periods. This decay stems from a combination of market learning (arbitrage capital exploiting newly disclosed anomalies) and data-mining biases in published literature.

2.3.2 Multiple Testing Inflation and False Discovery Rates

The main statistical cause of factor zoo inflation was analyzed by Harvey et al. (2016). Traditional empirical asset pricing evaluated candidate factors using single-hypothesis Student's t-tests with standard significance thresholds ($|t| > 1.96$, corresponding to $\alpha = 0.05$). However, when hundreds of candidate signals are evaluated on the same datasets, standard significance cutoffs produce high false discovery rates due to multiple hypothesis testing.

Harvey et al. (2016) evaluated over 300 published anomaly factors under multiple testing adjustments (including Bonferroni corrections and Holm-Bonferroni procedures). They showed that to maintain a true False Discovery Rate (FDR) below 5%, the minimum t-statistic cutoff for newly proposed asset pricing factors should be elevated to at least:

$$|t| > 3.0 \quad (p\text{-value} < 0.0027) \tag{2.10}$$

Extending this framework, Harvey and Liu (2020) introduced Bayesian FDR frameworks that account for prior belief distributions on true alpha. In a large replication audit, Hou et al. (2020) re-examined 452 cross-sectional anomalies under microcap exclusions and value-weighted portfolio rules. They showed that 65% of published anomalies fail to achieve statistical significance at standard $|t| > 1.96$ thresholds, and over 85% fail under the $|t| > 3.0$ threshold proposed by Harvey et al. (2016).

2.3.3 High-Dimensional Regularized SDF Estimation

To address the factor zoo problem structurally, econometricians re-framed asset pricing as a high dimensional regularized Stochastic Discount Factor (SDF) estimation problem. In a stochastic discount factor framework, an asset pricing model specifies a scalar pricing kernel M_{t+1} such that for all excess returns $R_{i,t+1}$:

$$\mathbb{E}_t\left[M_{t+1}R_{i,t+1}\right] = 0 \tag{2.11}$$

When the SDF is parameterized as a linear combination of a high dimensional vector of K candidate characteristics or factor returns $\mathbf{z}_{t+1}$, i.e., $M_{t+1} = 1 - \mathbf{b}^T(\mathbf{z}_{t+1} - \mathbb{E}[\mathbf{z}_{t+1}])$, unregularized sample estimators fail due to multicollinearity and $K \approx T$ sample noise.

Kozak et al. (2020) showed that shrinking the factor weighting vector $\mathbf{b}$ via combined L_1 (LASSO) and L_2 (Ridge) regularization yields robust, out-of-sample optimal SDF estimators. The regularized SDF objective function is formulated as:

$$\min_{\mathbf{b}} \left\{ \frac{1}{T} \sum_{t=1}^{T} \left(1 - \mathbf{b}^T\mathbf{z}_t\right)^2 + \gamma_1\|\mathbf{b}\|_1 + \gamma_2\|\mathbf{b}\|_2^2 \right\} \tag{2.12}$$

Kozak et al. (2020) showed that while unregularized models tend to overfit high-frequency sample noise, penalized SDF estimation extracts a low-dimensional principal component structure that spans expected returns while shrinking noise coefficients to zero.

2.4 High-Dimensional SDF Estimation & Characteristic Interactions

2.4.1 Non-Parametric Characteristic Pruning via Group LASSO

Parallel to regularized linear SDFs, Freyberger et al. (2020) developed a nonparametric model to systematically select which firm characteristics provide incremental predictive power for expected returns. Using nonparametric B-spline expansions and Group LASSO penalties, their estimator models expected returns as:

$$\mathbb{E}_t[R_{i,t+1}] = \sum_{j=1}^{J} g_j(c_{i,j,t}) + \epsilon_{i,t+1} \tag{2.13}$$

where $c_{i,j,t}$ represents the rank-transformed characteristic j for firm i at time t, and $g_j(\cdot)$ is a nonlinear continuous function. The Group LASSO penalty penalizes the vector of spline coefficients for each characteristic simultaneously:

$$\min_{\{g_j\}} \left\{ \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(R_{i,t+1} - \sum_{j=1}^{J} g_j(c_{i,j,t}) \right)^2 + \lambda \sum_{j=1}^{J} \sqrt{\text{dim}(g_j)} \|g_j\|_2 \right\} \tag{2.14}$$

Freyberger et al. (2020) showed that out of more than 50 published characteristics, only a small subset (such as short term momentum, market capitalization, operating profitability, and asset growth) retain independent predictive power once non-linearities and characteristic redundancies are penalized.

2.4.2 Sparse Signals and Latent Factor Shrinkage

Chinco et al. (2019) examined sparse return predictability at high frequencies, showing that LASSO regression identifies transient, cross-asset predictive signals that standard time series estimators miss. Grounding model selection in econometric diagnostics, Lewellen et al. (2010) provided a critique of cross-sectional asset pricing tests. They showed that testing multifactor models on strong factor-structure portfolios (such as standard 25 Size-B/M sorted portfolios) produces artificially inflated cross-sectional R^2 values. They prescribed rigorous diagnostic standards: expanding test portfolios beyond Size-B/M, forcing theoretical factor risk premia to equal sample factor means, and reporting confidence intervals for cross-sectional R^2.

Expanding structural factor selection, Feng et al. (2020) and Giglio and Xiu (2021) developed double-testing econometric frameworks combining LASSO factor selection with two-pass regressions. They showed that high dimensional regularization enables researchers to evaluate whether a newly proposed factor adds incremental explanatory power to the existing universe of 300+ factors without suffering from omitted variable bias or data mining contamination.

2.5 Machine Learning and Deep Learning Paradigms

2.5.1 Empirical Asset Pricing Benchmark: Gu, Kelly, and Xiu (2020)

The landmark study by Gu et al. (2020) established a primary theoretical and empirical benchmark for financial machine learning. Analyzing a panel of over 30,000 U.S. equities over 60 years (1957–2016) with 94 firm-level characteristics and 8 macroeconomic indicators, Gu, Kelly, and Xiu evaluated a range of machine learning architectures: Linear OLS, LASSO, ElasticNet, Principal Component Regression (PCR), Partial Least Squares (PLS), Random Forests, Gradient Boosted Decision Trees (GBDT), and multilayer Feedforward Neural Networks (MLP).

The nonlinear forecasting panel model is defined as:

$$R_{i,t+1} = g^*(\mathbf{x}_{i,t}) + \epsilon_{i,t+1}, \quad \mathbf{x}_{i,t} = \mathbf{c}_{i,t} \otimes \mathbf{d}_t \tag{2.15}$$

where $\mathbf{c}_{i,t}$ is a p-dimensional vector of firm characteristics, $\mathbf{d}_t$ is an m-dimensional vector of macro variables, and $\otimes$ denotes the Kronecker product creating interactive feature conditioning.

Gu et al. (2020) highlighted three main empirical findings:

1. **Non-Linear Model Performance**: Nonlinear models (GBDT, Neural Networks) outperform linear models in out-of-sample predictive R^2_{oos}, doubling the annualized Sharpe ratios of long-short hedge portfolios.

2. **Importance of Interaction Terms**: A main source of machine learning gain comes from automatically capturing nonlinear characteristic interactions (such as the interaction of size with short term momentum and liquidity).

3. **Neural Network Architecture**: Shallow neural architectures with 1 to 3 hidden layers achieve better performance compared to deeper networks, as excessive depth can lead to overfitting on low signal-to-noise equity return data.

2.5.2 Characteristics as Dynamic Risk Exposures: IPCA

Theoretical motivation for why machine learning return forecasts succeed was provided by Kelly et al. (2019). Introducing Instrumented Principal Component Analysis (IPCA), Kelly, Pruitt, and Su modeled asset returns as:

$$R_{i,t+1} = \mathbf{z}_{i,t}^T \mathbf{\Gamma}_\beta \mathbf{f}_{t+1} + \epsilon_{i,t+1} \tag{2.16}$$

where $\mathbf{z}_{i,t}$ is a vector of observable firm characteristics, $\mathbf{\Gamma}_\beta$ is a mapping matrix connecting characteristics to dynamic factor loadings ($\boldsymbol{\beta}_{i,t} = \mathbf{\Gamma}_\beta^T \mathbf{z}_{i,t}$), and $\mathbf{f}_{t+1}$ represents unobservable latent factor returns.

Kelly et al. (2019) showed that firm characteristics serve as proxies for time-varying factor risk exposures. Consequently, machine learning return predictors predicting $R_{i,t+1} = g(\mathbf{z}_{i,t})$ are implicitly estimating dynamic latent factor models with time-varying risk loadings.

2.5.3 Deep Learning and Non-Parametric SDF Estimation

Extending ML to advanced deep architectures, Chen et al. (2024) developed deep neural network models that estimate nonparametric asset pricing SDFs directly from high dimensional equity panels. By parameterizing factor loadings $\beta(\mathbf{c}_{i,t})$ as nonlinear functions of firm characteristics using Deep Neural Networks (DNNs) combined with Generative Adversarial Networks (GANs), Chen, Pelger, and Zhu showed that deep learning captures macroeconomic regime shifts and structural non-linearities that linear factor models may miss.

Similarly, Cong et al. (2021) incorporated multimodal deep learning architectures to process accounting signals alongside high dimensional textual information. To help explain the predictions of deep neural networks, Lundberg and Lee (2017) introduced

SHapley Additive exPlanations (SHAP), based on cooperative game theory. SHAP decomposes a model prediction $f(\mathbf{x})$ into additive feature contributions:

$$f(\mathbf{x}) = \phi_0 + \sum_{j=1}^{M} \phi_j \mathbf{x}_j \tag{2.17}$$

providing transparent economic interpretation of nonlinear asset pricing models.

2.5.4 Economic Frictions and Limits to Arbitrage

While machine learning models achieve notable statistical R^2_{oos}, Avramov et al. (2023) investigated whether these return forecast gains survive real-world economic frictions. Avramov, Cheng, and Metzker showed that machine learning return predictability is heavily concentrated in illiquid, microcap stocks with high distress risk, high transaction costs, and borrowing constraints. When portfolios exclude microcaps or account for realistic execution costs, out-of-sample profits decline significantly. This highlights the importance of integrating microstructural frictions into financial machine learning evaluation.

2.6 Financial Time-Series Econometrics & Inference

2.6.1 Heteroskedasticity and Autocorrelation Consistent (HAC) Inference

Financial return time series exhibit conditional heteroskedasticity, volatility clustering, and serial dependence. Standard Ordinary Least Squares (OLS) covariance estimators understate parameter variances, yielding artificially inflated t-statistics. To provide valid statistical inference, Newey and West (1987) developed the Heteroskedasticity and Autocorrelation Consistent (HAC) covariance matrix estimator.

For a time series of regression residuals e_t and regressors $\mathbf{X}_t$, defining $\mathbf{u}_t = \mathbf{X}_t e_t$, the Newey-West estimator computes the sample covariance matrix $\widehat{\Omega}_{\text{NW}}$ as:

$$\widehat{\Omega}_{\text{NW}} = \widehat{\Gamma}_0 + \sum_{j=1}^{L} w(j, L) \left(\widehat{\Gamma}_j + \widehat{\Gamma}_j^T \right) \tag{2.18}$$

where $\widehat{\Gamma}_j = \frac{1}{T} \sum_{t=j+1}^{T} \mathbf{u}_t \mathbf{u}_{t-j}^T$ is the sample autocovariance matrix at lag j, and $w(j, L) = 1 - \frac{j}{L+1}$ represents the Bartlett kernel weighting function over a maximum lag length L. The Bartlett kernel ensures that $\widehat{\Omega}_{\text{NW}}$ is positive semi-definite in finite samples.

2.6.2 Out-of-Sample Predictive Accuracy: Diebold-Mariano and HLN Corrections

When comparing out-of-sample forecast accuracy between two competing forecasting models (such as a deep sequence network versus a baseline linear factor model), standard independent sample t-tests are invalid due to forecast error autocorrelation. Diebold and Mariano (1995) introduced a distribution-free statistic for testing the null hypothesis of equal predictive accuracy.

Let $e_{1,t}$ and $e_{2,t}$ denote forecast errors from Model 1 and Model 2 at time t. The loss differential sequence is defined under a loss function $L(\cdot)$ (such as Mean Squared Error):

$$d_t = L(e_{1,t}) - L(e_{2,t}) \tag{2.19}$$

The null hypothesis of equal forecast accuracy is $H_0 : \mathbb{E}[d_t] = 0$. The Diebold-Mariano (DM) test statistic is given by:

$$DM = \frac{\bar{d}}{\sqrt{\frac{\widehat{V}(\bar{d})}{T}}} \xrightarrow{d} \mathcal{N}(0, 1) \tag{2.20}$$

where $\bar{d} = \frac{1}{T}\sum_{t=1}^{T} d_t$, and $\widehat{V}(\bar{d})$ is the HAC-adjusted asymptotic variance of the loss differential sequence using the Newey-West estimator.

To correct for small-sample size distortion in modest evaluation panels, Harvey et al. (1997) introduced the Harvey-Leybourne-Newbold (HLN) small-sample correction. For a forecast horizon h, the HLN-adjusted statistic is:

$$DM^* = DM \times \left[\frac{T + 1 - 2h + T^{-1}h(h-1)}{T}\right]^{1/2} \tag{2.21}$$

Critical values for DM^* are drawn from the Student's t-distribution with $(T-1)$ degrees of freedom, providing robust inference in empirical financial panels.

2.6.3 Data Mining Diagnostics: Reality Check and Superior Predictive Ability

When searching across multiple model hyperparameter configurations or architecture variants, researchers risk selecting a model that performs well out-of-sample purely by chance. To control for data mining, White (2000) introduced the Reality Check for Data Mining. White formulated a bootstrap procedure over the joint distribution of benchmark and candidate model forecasts to test whether the best performing candidate model genuinely outperforms the benchmark.

Recognizing that White's Reality Check can be conservative when poor candidate models are included, Hansen (2005) refined the framework into the Superior Predictive Ability (SPA) test. Hansen introduced a studentized test statistic with sample-dependent variance adjustments:

$$T_k^{\text{SPA}} = \max_{k=1,\dots,K} \frac{\sqrt{T}\bar{d}_k}{\hat{\omega}_k} \tag{2.22}$$

where $\hat{\omega}_k^2$ is a consistent estimator of the asymptotic variance of $\sqrt{T}\bar{d}_k$. Hansen showed that the SPA test exhibits higher power and robustness against uninformative candidate models, serving as a useful diagnostic for quantitative model comparison.

2.7 Financial ML Protocols & Overfitting Mitigation

2.7.1 Methodological Pitfalls in Financial Machine Learning

In financial machine learning, standard statistical cross-validation techniques (such as standard randomized K-fold CV) can fail. As detailed by López de Prado (2018), financial time series data violate the independent and identically distributed (i.i.d.) assumption. Applying naive cross-validation induces several methodological problems:

1. **Lookahead Leakage**: Using future information (such as future target returns or lookahead normalized features) during training.

2. **Serial Memory Contamination**: Overlapping target return windows (such as multi-day cumulative returns) allow training samples to overlap with test samples, leaking labels across folds.

3. **Backtest Overfitting**: Iteratively tuning model parameters directly on test set Sharpe ratios until an illusion of profitability is produced.

2.7.2 Validation Protocols

To ensure backtesting integrity, López de Prado (2018) established validation protocols:

- **Expanding Walk-Forward Validation**: Models are trained on historical data $t \in [1, T_{\text{train}}]$ and evaluated on subsequent out-of-sample periods $t \in [T_{\text{train}} + 1, T_{\text{test}}]$. The training window expands iteratively over time.
- **Combinatorial Purged Cross-Validation (CPCV)**: When non-sequential cross-validation is conducted, training folds adjacent to test folds are *purged* by removing samples whose label windows overlap with test windows. In addition, an *embargo* period is appended after test sets to eliminate serial correlation leakage.
- **Point-in-Time Data Alignment**: Financial fundamentals and accounting metrics are indexed by their actual public SEC disclosure dates (such as 10-K/10-Q filing dates) rather than fiscal period end dates.

2.7.3 Microstructure Frictions & Transaction Cost Taxonomy

High-frequency predictive signals can decay rapidly once realistic transaction costs and market microstructure frictions are accounted for (Novy-Marx and Velikov, 2016; Frazzini and Pedersen, 2014). Novy-Marx and Velikov (2016) show that many published anomaly strategies disappear after deducting effective bid-ask spreads, market impact costs, and turnover frictions.

Second, execution realism mandates an execution delay: trading signals generated at market close on day t cannot be executed until day $t + 1$ close or open. In addition, leverage constraints and margin costs (Frazzini and Pedersen, 2014) restrict short-selling capacity on distressed equities, reinforcing the necessity of evaluating long-only and cost-adjusted long-short performance.

2.8 Synthesis of Core Literature Benchmarks

Table 2.1 provides a synthesis of the literature that anchors the theoretical framing, model architecture, feature engineering, econometric estimation, and validation of this thesis.

Table 2.1: Comprehensive Synthesis of Core Literature Benchmarks

Thematic Domain	Key Literature Citation	Core Theoretical / Econometric Finding	Implementation in Thesis
Asset Pricing Foundations	Sharpe (1964), Lintner (1965)	Single factor CAPM equilibrium; market beta linear pricing	Baseline risk benchmark
	Fama and MacBeth (1973)	Two-pass cross-sectional regression methodology	Baseline linear factor risk premia
	Ross (1976)	Arbitrage Pricing Theory (APT); multifactor space	Linear multifactor baseline
	Fama and French (1993), Fama and French (2015)	3-Factor and 5-Factor models (Mkt, SMB, HML, RMW, CMA)	Econometric spanning regressions
	Carhart (1997), Pástor and Stambaugh (2003)	Price momentum (WML) and aggregate market liquidity factors	Extended multifactor benchmark
Factor Zoo & Shrinkage	Cochrane (2011)	Factor zoo diagnosis; lack of unified factor structure	High dimensional model motivation
	Harvey et al. (2016), Harvey and Liu (2020)	Multiple testing bias; minimum hurdle rate $\|t\| > 3.0$	Feature auditing threshold
	Hou et al. (2020), McLean and Pontiff (2016)	65%+ anomaly replication failure; post-publication decay	Microcap exclusions & value weighting
	Kozak et al. (2020)	Regularized L_1/L_2 SDF shrinkage in high dimensions	Penalized Ridge/ElasticNet baselines
	Freyberger et al. (2020)	Nonparametric Group LASSO characteristic pruning	Rank-normalized feature selection
Machine Learning & Deep Learning	Gu et al. (2020)	Nonlinear ML performance; interaction terms; 1–3 layer networks	Experimental lineup design
	Kelly et al. (2019)	Instrumented PCA (IPCA); characteristics as dynamic betas	Latent dynamic factor framing
	Chen et al. (2024)	Deep learning nonparametric SDF estimation via GANs	Sequence model architecture
	Avramov et al. (2023)	ML gains concentrated in illiquid hard-to-arbitrage stocks	Size-stratified empirical evaluation
	Lundberg and Lee (2017)	SHAP additive feature attribution for nonlinear models	Model interpretability framework
Time-Series Econometrics	Newey and West (1987)	HAC robust standard errors; Bartlett kernel weighting	Fama-MacBeth HAC t-statistics
	Diebold and Mariano (1995), Harvey et al. (1997)	Out-of-sample forecast accuracy test & HLN correction	Model forecast comparison tests
	White (2000), Hansen (2005)	Reality Check and Superior Predictive Ability (SPA) tests	Hyperparameter data-mining audit
Financial ML Validation	López de Prado (2018)	Financial ML protocols; purging, embargoing, walk-forward	5-Fold expanding walk-forward split
	Novy-Marx and Velikov (2016)	Transaction cost taxonomy; effective spreads and impact	10 bps friction drag evaluation
	Frazzini and Pedersen (2014)	Betting against beta; leverage constraints and margin drag	Execution delay & shorting rules

2.9 Summary

In conclusion, the evolution of empirical asset pricing shows a clear transition from simple linear factor models toward regularized high dimensional estimation and nonparametric deep sequence architectures. By synthesizing the theoretical insights of factor pricing, the statistical rigor of multiple testing adjustments, the predictive capability of deep sequence networks, and empirical validation protocols, this thesis builds a framework for cross-sectional stock return prediction.

3. Data & Methodology

3.1 Data Pipeline and Sample Construction

My empirical sample comprises daily stock prices, trading volumes, and quarterly financial accounting metrics for active and historical S&P 500 constituent equities from December 31, 2014, through December 31, 2024. The active evaluation window spans January 1, 2015, to December 31, 2024, yielding 2,516 trading days and over 1.25 million stock-day observations (Gu et al., 2020).

Daily closing prices, dividend payments, stock splits, and trading volumes are ingested from Yahoo Finance and cross-verified against Bloomberg Terminal records. Daily risk-free rates ($R_{f,t}$) along with daily Fama-French 5-Factor returns ($Mkt - RF_t, SMB_t, HML_t, RMW_t, CMA_t$) and Carhart Momentum ($MOM_t$) are retrieved directly from the Ken French Data Library (Fama and French, 1993, 2015; Carhart, 1997).

3.1.1 Point-in-Time Bloomberg Filing Date Alignment

A common issue in empirical finance studies is lookahead bias caused by indexing corporate accounting metrics to fiscal period end-dates. In practice, financial statements for a fiscal quarter ending on December 31 (t_{fiscal}) are published weeks or months later during the 10-K or 10-Q SEC filing process (López de Prado, 2018).

To eliminate lookahead contamination, my data pipeline (`data_pipeline.py` in `src/`) ingests official Bloomberg publication timestamps (`earnings_announcement_date` and `filing_date`). On any given trading date t, a quarterly fundamental feature vector $\mathbf{x}_{i,t}^{\text{fund}}$ is updated to the newly reported accounting state Q_τ if and only if:

$$t \geq \text{Announcement Date}(Q_\tau) \tag{3.1}$$

Prior to $t = \text{Announcement Date}(Q_\tau)$, the model is strictly limited to using the previously announced fundamental vector $Q_{\tau-1}$.

In addition, I enforce a **Staleness Truncation Rule**: a fundamental observation remains active for at most 90 trading days (≈ 1 calendar quarter). If a firm fails to publish a new statement within 120 calendar days (≈ 90 trading days), the forward-fill is truncated and the corresponding fundamental fields are set to missing (`NaN`), preventing outdated accounting ratios from skewing cross-sectional ranks (Novy-Marx and Velikov, 2016).

3.1.2 Survivorship-Bias-Free Dynamic Panel

To guard against survivorship bias, my asset universe tracks historical point-in-time S&P 500 constituent rosters obtained from Bloomberg. Companies removed from the index, acquired, or delisted due to bankruptcy between 2015 and 2024 remain in the daily cross-section up to their final active trading date T_{exit} (McLean and Pontiff, 2016; Hou et al., 2020).

Figure 3.1 shows my point-in-time data alignment pipeline and staleness truncation flowchart.

Figure 3.1: Bloomberg Point-in-Time Data Pipeline and Staleness Truncation Flowchart

3.2 Feature Space Taxonomy and Selection

My empirical panel relies on **59 variables in total**, split between predictive model inputs and benchmark risk factors:

- **53 Predictive Machine Learning Inputs**: Fed directly into the machine learning and neural network architectures across four modalities: Technical (16), Fundamental (30), Sentiment (2), and Macro (5) (Gu et al., 2020; Cong et al., 2021).
- **6 Benchmark Risk Factor Betas**: Retained for baseline linear regressions and econometric risk adjustments, including the 5 Fama-French systematic factor betas ($Mkt - RF, SMB, HML, RMW, CMA$) and Momentum ($MOM$) (Fama and French, 2015; Carhart, 1997).

3.2.1 Economic Rationale for Feature Set Size (59 Variables vs. 120+ Zoo Features)

When designing the characteristic space, a key decision is whether to include hundreds of candidate anomaly signals or focus on a curated panel. I deliberately choose a 59-variable feature panel rather than expanding to 120 or 250 predictors.

Including vast numbers of firm characteristics in financial panels introduces practical and statistical problems. Many raw accounting metrics simply restate similar economic signals, such as minor variations of leverage ratios or momentum lookbacks, adding collinearity and noise rather than genuine predictive content (Freyberger et al., 2020; Kozak et al., 2020). In addition, as Harvey et al. (2016) observe, testing hundreds of features increases false-discovery rates. Restricting the predictor set to 53 ML inputs grounded in four economic domains (Technical, Fundamental, Sentiment, and Macro) ensures each variable represents an established economic mechanism (Pástor and Stambaugh, 2003; Novy-Marx, 2013; Cong et al., 2021; Fama and French, 2015). Finally, requiring hundreds of characteristics reduces cross-sectional sample coverage, as missing fundamental items force the exclusion of valid stock-day observations (Avramov et al., 2023).

Figure 3.2 displays the feature group modality classification.

Figure 3.2: Variable Taxonomy and Feature Group Modality Split (53 ML Inputs + 6 Factor Betas = 59 Total)

3.3 Feature Normalization and Selection Protocol

To maintain uniform scaling and stationarity, raw characteristic variables are converted each day into **Cross-Sectional Ranks** mapped to the unit interval $[0, 1]$ (Freyberger et al., 2020; Gu et al., 2020):

$$r_{i,t,j} = \frac{\mathrm{Rank}(X_{i,t,j}) - 1}{N_t - 1} \in [0, 1] \tag{3.2}$$

Rank-scaling protects models from extreme financial outliers, standardizes variables across different accounting units, and keeps distributions comparable across changing market environments.

Importantly, cross-sectional rank normalization in Equation (3.2) is applied strictly to firm-level characteristics (Technical and Fundamental features). Macroeconomic state variables (x_{macro}) and news/social sentiment indicators are **not** rank-normalized cross-sectionally across stocks on day t — doing so would reduce their cross-sectional variance to zero, as macro variables are uniform across all assets on a given trading day. Instead, macro state variables are normalized temporally across time or processed directly by the 3-Way Macro Dynamic Gating network.

To guarantee zero lookahead leakage and zero overnight gap contamination, return targets $y_{i,t+1}$ in training and evaluation are strictly computed from $P_{i,t+1,\text{open}}$ to $P_{i,t+2,\text{open}}$ (or $P_{i,t+1,\text{close}}$ to $P_{i,t+2,\text{close}}$ following execution at $t + 1$), matching the 1-day execution buffer delay (shift(-2)).

To avoid data leakage, I perform feature selection (`select_features.py` in `scripts/`) using a **three-stage protocol applied exclusively to pre-2020 training data** (146,959 observations) (López de Prado, 2018). The first stage removes features with missing rates above 10%. The second stage prunes redundant variables exhibiting pairwise Spearman correlations $|\rho_{j,k}| > 0.85$. In the third stage, an in-sample Random Forest regressor (Breiman, 2001) ranks remaining candidate variables, selecting the top 53 predictive features.

Figure 3.3 presents the correlation matrix of selected features, and Figure 3.4 shows the Random Forest importance rankings.

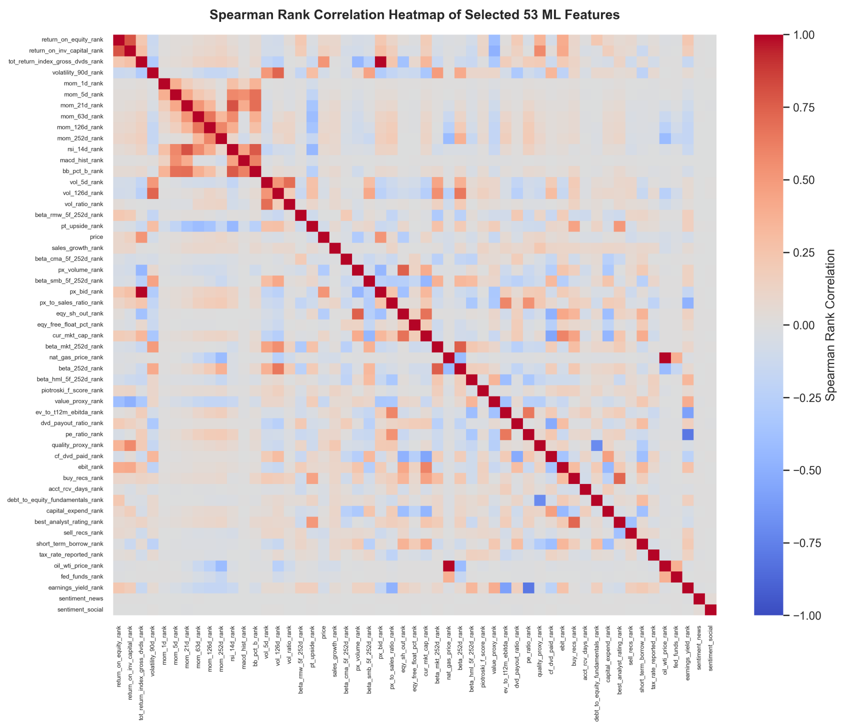

Figure 3.3: Spearman Rank Correlation Heatmap of Selected 53 ML Features

Figure 3.4 reports the in-sample permutation feature importances calculated strictly on pre-2020 training data to prevent out-of-sample lookahead leakage. A classic methodological issue in tree-based feature selection is that raw Gini impurity measures over-attribute importance to continuous, un-normalized variables (Strobl et al., 2007). Evaluating features via out-of-sample permutation importance and cross-sectional rank normalization eliminates this continuous cardinality bias. Technical momentum indicators (vol_{126d}, mom_{21d}, rsi_{14d}) and fundamental metrics (pe_{ratio}, roi, quality) emerge as

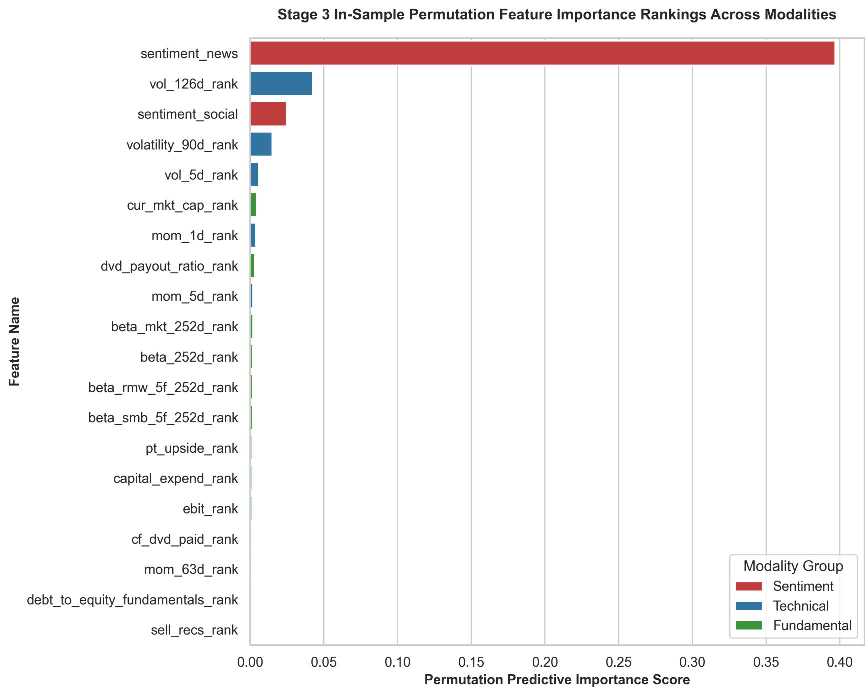

Figure 3.4: Stage 3 In-Sample Permutation Feature Importance Rankings across Modalities

the dominant predictive drivers, while sentiment indicators provide moderate secondary refinement, aligning fully with downstream SHAP values (Figure 4.2) and modality ablation evidence (Table 4.3).

3.4 Baseline Econometric & Machine Learning Formulations

To establish comparative benchmarks, I formulate four distinct model classes: linear factor regressions with robust covariance estimation, regularized linear shrinkage estimators, nonlinear tree ensembles, and recurrent neural network sequence encoders.

3.4.1 Fama-MacBeth 2-Pass Linear Regression & Newey-West HAC Estimation

The benchmark linear asset pricing model follows the two-pass cross-sectional regression procedure of Fama and MacBeth (1973), incorporating 1-day lagged factor loadings to prevent lookahead bias and heteroskedasticity and autocorrelation consistent (HAC) covariance matrices (Newey and West, 1987).

Pass 1: Rolling Time-Series Factor Loading Estimation

For each individual stock $i \in \{1, \ldots, N_t\}$ on trading day t, factor loadings (betas) are estimated over a rolling historical lookback window of $W = 252$ trading days (one calendar year). Let $\mathbf{R}_{i,t} = [R_{i,t-W+1}, R_{i,t-W+2}, \ldots, R_{i,t}]^T \in \mathbb{R}^{W\times 1}$ denote the vector of historical daily excess returns for asset i, and let $\mathbf{F}_t \in \mathbb{R}^{W\times(K+1)}$ represent the matrix of K benchmark factor returns plus a column of ones for the intercept:

$$\mathbf{F}_t = \begin{bmatrix} 1 & F_{1,\,t-W+1} & F_{2,\,t-W+1} & \cdots & F_{K,\,t-W+1} \\ 1 & F_{1,\,t-W+2} & F_{2,\,t-W+2} & \cdots & F_{K,\,t-W+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & F_{1,\,t} & F_{2,\,t} & \cdots & F_{K,\,t} \end{bmatrix} \tag{3.3}$$

The first-pass time series Ordinary Least Squares (OLS) regression for stock i is formulated as:

$$\mathbf{R}_{i,t} = \mathbf{F}_t \boldsymbol{\beta}_{i,t} + \boldsymbol{\epsilon}_{i,t} \tag{3.4}$$

where $\boldsymbol{\beta}_{i,t} = [\alpha_{i,t}, \beta_{i,1,t}, \beta_{i,2,t}, \ldots, \beta_{i,K,t}]^T \in \mathbb{R}^{(K+1)\times 1}$. The OLS point estimator is obtained by minimizing the sum of squared residuals:

$$\hat{\boldsymbol{\beta}}_{i,t} = \arg\min_{\boldsymbol{\beta}} (\mathbf{R}_{i,t} - \mathbf{F}_t\boldsymbol{\beta})^T(\mathbf{R}_{i,t} - \mathbf{F}_t\boldsymbol{\beta}) = \left(\mathbf{F}_t^T\mathbf{F}_t\right)^{-1}\mathbf{F}_t^T\mathbf{R}_{i,t} \tag{3.5}$$

To enforce zero-lookahead protection, factor loadings $\hat{\boldsymbol{\beta}}_{i,t}$ estimated using data up to day t are indexed to predict returns at day $t+1$.

Pass 2: Daily Cross-Sectional Factor Risk Premia Estimation

On each trading day $t + 1$, a cross-sectional OLS regression is estimated across all active constituent assets N_{t+1} using the 1-day lagged factor loadings $\hat{\mathbf{B}}_t = [\mathbf{1}, \hat{\boldsymbol{\beta}}_{1,t}, \hat{\boldsymbol{\beta}}_{2,t}, \ldots, \hat{\boldsymbol{\beta}}_{N_{t+1},t}]^T \in \mathbb{R}^{N_{t+1} \times (K+1)}$:

$$\mathbf{R}_{t+1} = \hat{\mathbf{B}}_t \boldsymbol{\gamma}_{t+1} + \boldsymbol{\eta}_{t+1} \tag{3.6}$$

where $\mathbf{R}_{t+1} = [R_{1,t+1}, R_{2,t+1}, \ldots, R_{N_{t+1},t+1}]^T \in \mathbb{R}^{N_{t+1} \times 1}$ is the cross-section of realized forward returns, $\boldsymbol{\gamma}_{t+1} = [\gamma_{0,t+1}, \gamma_{1,t+1}, \ldots, \gamma_{K,t+1}]^T \in \mathbb{R}^{(K+1) \times 1}$ represents the vector of daily factor risk premia (and cross-sectional intercept $\gamma_{0,t+1}$), and $\boldsymbol{\eta}_{t+1}$ is the vector of cross-sectional pricing errors.

The closed-form daily cross-sectional risk premium estimator is given by:

$$\hat{\gamma}_{t+1} = \left(\hat{\mathbf{B}}_t^T \hat{\mathbf{B}}_t\right)^{-1} \hat{\mathbf{B}}_t^T \mathbf{R}_{t+1} \tag{3.7}$$

Time-Series Aggregation & Step-by-Step Newey-West HAC Proof

Over an evaluation period of T trading days, the overall factor risk premium vector is estimated as the sample mean across daily cross-sectional estimates:

$$\bar{\gamma} = \frac{1}{T} \sum_{t=1}^{T} \hat{\gamma}_t \tag{3.8}$$

Because daily factor risk premia estimates $\hat{\gamma}_t$ suffer from serial correlation (autocorrelation) due to overlapping return measurement horizons and conditional heteroskedasticity in asset volatility, standard i.i.d. variance estimators underestimate true standard errors, inflating test statistics.

To derive a consistent estimator for the asymptotic covariance matrix of $\bar{\boldsymbol{\gamma}}$, let $\mathbf{u}_t = \hat{\boldsymbol{\gamma}}_t - \bar{\boldsymbol{\gamma}} \in \mathbb{R}^{(K+1)\times 1}$ denote the vector of mean-zero risk premium innovations. The variance of the sample average factor risk premium is:

$$\mathrm{Var}(\bar{\boldsymbol{\gamma}}) = \mathrm{Var}\left(\frac{1}{T}\sum_{t=1}^{T}\hat{\boldsymbol{\gamma}}_t\right) = \frac{1}{T^2}\sum_{t=1}^{T}\sum_{s=1}^{T}\mathbb{E}\left[\mathbf{u}_t\mathbf{u}_s^T\right] \tag{3.9}$$

Defining the sample autocovariance matrix at lag j as:

$$\hat{\boldsymbol{\Gamma}}_j = \frac{1}{T}\sum_{t=j+1}^{T}\mathbf{u}_t\mathbf{u}_{t-j}^T, \quad \text{for } j \geq 0 \tag{3.10}$$

Equation (3.9) can be expanded into the sum of zero-lag covariance and off-diagonal cross-lag covariances:

$$\mathrm{Var}(\bar{\boldsymbol{\gamma}}) = \frac{1}{T}\left[\hat{\boldsymbol{\Gamma}}_0 + \sum_{j=1}^{T-1}\left(\hat{\boldsymbol{\Gamma}}_j + \hat{\boldsymbol{\Gamma}}_j^T\right)\right] \tag{3.11}$$

Unweighted truncation of lag covariances at finite lag $L < T$ can lead to non-positive semi-definite covariance matrices in finite samples. The Newey and West (1987) HAC covariance matrix estimator guarantees positive semi-definiteness by weighting lag autocovariances with the triangular Bartlett kernel $w(j, L)$:

$$\widehat{\boldsymbol{\Omega}}_{\text{HAC}} = \hat{\boldsymbol{\Gamma}}_0 + \sum_{j=1}^{L} w(j, L)\left(\hat{\boldsymbol{\Gamma}}_j + \hat{\boldsymbol{\Gamma}}_j^T\right) \tag{3.12}$$

where the Bartlett kernel weights are defined as:

$$w(j, L) = 1 - \frac{j}{L+1}, \quad j = 1, 2, \dots, L \tag{3.13}$$

and the lag truncation bandwidth L is set according to the bandwidth selection rule $L = \lfloor 4(T/100)^{2/9} \rfloor$.

Proof of Guaranteed Positive Semi-Definiteness. To prove that $\widehat{\boldsymbol{\Omega}}_{\text{HAC}} \succeq 0$, express the Bartlett kernel as the spectral window resulting from the convolution of a rectangular sequence with itself. Define $\mathbf{u}_t = \mathbf{0}$ for $t \notin \{1, \ldots, T\}$. Consider the smoothed sequence $\mathbf{s}_t = \frac{1}{\sqrt{L+1}} \sum_{m=0}^{L} \mathbf{u}_{t+m}$. The sample covariance of $\mathbf{s}_t$ is positive semi-definite:

$$\mathbf{M} = \frac{1}{T} \sum_{t=-\infty}^{\infty} \mathbf{s}_t \mathbf{s}_t^T = \frac{1}{T(L+1)} \sum_{t=-\infty}^{\infty} \left(\sum_{m=0}^{L} \mathbf{u}_{t+m} \right) \left(\sum_{n=0}^{L} \mathbf{u}_{t+n}^T \right) \geq 0 \tag{3.14}$$

Expanding the double summation yields:

$$\mathbf{M} = \frac{1}{T(L+1)} \sum_{m=0}^{L} \sum_{n=0}^{L} \sum_{t=-\infty}^{\infty} \mathbf{u}_{t+m} \mathbf{u}_{t+n}^T = \frac{1}{T(L+1)} \sum_{m=0}^{L} \sum_{n=0}^{L} T \hat{\mathbf{\Gamma}}_{m-n} \tag{3.15}$$

$$= \frac{1}{L+1} \sum_{k=-L}^{L} (L+1-|k|) \hat{\mathbf{\Gamma}}_k = \hat{\mathbf{\Gamma}}_0 + \sum_{j=1}^{L} \left(1 - \frac{j}{L+1}\right) \left(\hat{\mathbf{\Gamma}}_j + \hat{\mathbf{\Gamma}}_j^T\right) = \widehat{\mathbf{\Omega}}_{\text{HAC}} \tag{3.16}$$

Since $\mathbf{M}$ is constructed as a sum of outer products $\mathbf{s}_t \mathbf{s}_t^T$, for any vector $\mathbf{a} \in \mathbb{R}^{K+1}$, $\mathbf{a}^T \widehat{\mathbf{\Omega}}_{\text{HAC}} \mathbf{a} = \frac{1}{T} \sum_t (\mathbf{a}^T \mathbf{s}_t)^2 \geq 0$. Thus, $\widehat{\mathbf{\Omega}}_{\text{HAC}}$ is positive semi-definite. $\square$

The Newey-West adjusted standard error for the k-th factor risk premium estimate $\bar{\gamma}_k$ is:

$$\text{SE}_{\text{NW}}(\bar{\gamma}_k) = \sqrt{\frac{1}{T} \left[\widehat{\mathbf{\Omega}}_{\text{HAC}}\right]_{kk}} \tag{3.17}$$

and the test statistic for factor significance testing is evaluated as $t(\bar{\gamma}_k) = \frac{\bar{\gamma}_k}{\text{SE}_{\text{NW}}(\bar{\gamma}_k)}$.

3.4.2 Regularized Linear Models: ElasticNet, LASSO, & Ridge

High dimensional cross-sectional asset pricing settings with P characteristic features suffer from multicollinearity and sample overfitting under standard OLS. Regularized linear models introduce penalty structures to constrain coefficient magnitudes and perform feature selection (Kozak et al., 2020).

ElasticNet Objective Function

Let $\mathbf{y} \in \mathbb{R}^N$ denote the vector of standardized forward returns across all asset-day observations, and let $\mathbf{X} \in \mathbb{R}^{N \times P}$ denote the matrix of standardized stock characteristics. The ElasticNet estimator (Zou and Hastie, 2005) solves the convex optimization problem:

$$\min_{\boldsymbol{\beta} \in \mathbb{R}^P} \mathcal{L}_{\text{EN}}(\boldsymbol{\beta}) = \frac{1}{2N} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda \left[\alpha \|\boldsymbol{\beta}\|_1 + \frac{1-\alpha}{2} \|\boldsymbol{\beta}\|_2^2 \right] \tag{3.18}$$

where $\lambda > 0$ controls the penalty intensity, $\alpha \in [0, 1]$ governs the ratio between the L_1 norm $\|\boldsymbol{\beta}\|_1 = \sum_{j=1}^P |\beta_j|$ and the squared L_2 norm $\|\boldsymbol{\beta}\|_2^2 = \sum_{j=1}^P \beta_j^2$. Setting $\alpha = 1.0$ yields the LASSO estimator (Tibshirani, 1996), while $\alpha = 0.0$ yields Ridge regression (Hoerl and Kennard, 1970).

Subgradient Calculus & Characterization of LASSO (L_1) Sparsity

Because the L_1 norm is non-differentiable at $\beta_j = 0$, optimization requires subgradient calculus. The subdifferential $\partial|\beta_j|$ of the absolute value function is defined as:

$$\partial|\beta_j| = \begin{cases} \{+1\} & \text{if } \beta_j > 0 \\ \{-1\} & \text{if } \beta_j < 0 \\ [-1, 1] & \text{if } \beta_j = 0 \end{cases} \tag{3.19}$$

To derive the coordinate descent update rule for feature j, assume that the columns of $\mathbf{X}$ are standardized such that $\mathbf{x}_j^T \mathbf{x}_j = N$. Isolate feature j by defining the partial residual vector $\mathbf{r}^{(j)} = \mathbf{y} - \sum_{k \neq j} \mathbf{x}_k \beta_k$. The unregularized OLS update for β_j given fixed $\boldsymbol{\beta}_{-j}$ is $\hat{\beta}_j^{\text{OLS}} = \frac{1}{N} \mathbf{x}_j^T \mathbf{r}^{(j)}$.

The first order optimality condition for the ElasticNet objective requires $0 \in \partial_{\beta_j} \mathcal{L}_{\text{EN}}(\boldsymbol{\beta})$:

$$-\frac{1}{N} \mathbf{x}_j^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda(1-\alpha)\beta_j + \lambda\alpha v_j = 0, \quad \text{where } v_j \in \partial|\beta_j| \tag{3.20}$$

Substituting $\mathbf{y} - \mathbf{X}\boldsymbol{\beta} = \mathbf{r}^{(j)} - \mathbf{x}_j \beta_j$ and using $\frac{1}{N} \mathbf{x}_j^T \mathbf{x}_j = 1$:

$$-\hat{\beta}_j^{\text{OLS}} + \beta_j + \lambda(1-\alpha)\beta_j + \lambda\alpha v_j = 0 \implies \beta_j[1 + \lambda(1-\alpha)] + \lambda\alpha v_j = \hat{\beta}_j^{\text{OLS}} \tag{3.21}$$

Solving Equation (3.21) under three case conditions for $\hat{\beta}_j^{\text{OLS}}$:

1. **Case 1:** $\beta_j > 0$. Here $v_j = +1$. Equation (3.21) gives $\beta_j[1 + \lambda(1-\alpha)] = \hat{\beta}_j^{\text{OLS}} - \lambda\alpha$, requiring $\hat{\beta}_j^{\text{OLS}} > \lambda\alpha$. Thus:

$$\beta_j^* = \frac{\hat{\beta}_j^{\text{OLS}} - \lambda\alpha}{1 + \lambda(1-\alpha)}$$

2. **Case 2:** $\beta_j < 0$. Here $v_j = -1$. Equation (3.21) gives $\beta_j[1 + \lambda(1-\alpha)] = \hat{\beta}_j^{\text{OLS}} + \lambda\alpha$, requiring $\hat{\beta}_j^{\text{OLS}} < -\lambda\alpha$. Thus:

$$\beta_j^* = \frac{\hat{\beta}_j^{\text{OLS}} + \lambda\alpha}{1 + \lambda(1-\alpha)}$$

3. **Case 3:** $\beta_j = 0$. Here $v_j \in [-1, 1]$. Equation (3.21) requires $|\hat{\beta}_j^{\text{OLS}}| \leq \lambda\alpha$. Thus, $\beta_j^* = 0$.

Combining these cases yields the closed-form Soft-Thresholding Operator $\mathcal{S}_\kappa(z) = \text{sign}(z) \max(0, |z| - \kappa)$:

$$\hat{\beta}_j^{\text{EN}} = \frac{\mathcal{S}_{\lambda\alpha}\left(\hat{\beta}_j^{\text{OLS}}\right)}{1 + \lambda(1-\alpha)} \tag{3.22}$$

Setting $\alpha = 1.0$ yields the standard LASSO soft-thresholded estimator $\hat{\beta}_j^{\text{LASSO}} = \mathcal{S}_\lambda\left(\hat{\beta}_j^{\text{OLS}}\right)$, which sets coefficients to zero whenever $|\hat{\beta}_j^{\text{OLS}}| \leq \lambda$, performing feature selection.

Ridge (L_2) Shrinkage Bounds & SVD Derivation

For Ridge regression ($\alpha = 0$), the objective function is smooth and differentiable. Differentiating $\mathcal{L}_{\text{Ridge}}(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$ yields the closed-form Ridge estimator:

$$\hat{\boldsymbol{\beta}}_{\text{Ridge}} = \left(\mathbf{X}^T\mathbf{X} + N\lambda\mathbf{I}_P\right)^{-1}\mathbf{X}^T\mathbf{y} \tag{3.23}$$

To derive L_2 shrinkage bounds, consider the Singular Value Decomposition (SVD) of the centered design matrix $\mathbf{X} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^T$, where $\mathbf{U} \in \mathbb{R}^{N\times P}$ and $\mathbf{V} \in \mathbb{R}^{P\times P}$ are orthonormal matrices ($\mathbf{U}^T\mathbf{U} = \mathbf{I}_P$, $\mathbf{V}^T\mathbf{V} = \mathbf{I}_P$), and $\boldsymbol{\Sigma} = \text{diag}(\sigma_1, \sigma_2, \ldots, \sigma_P)$ contains the singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_P > 0$.

Substituting the SVD into the standard OLS estimator $\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$:

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = \left(\mathbf{V}\boldsymbol{\Sigma}^2\mathbf{V}^T\right)^{-1}\mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^T\mathbf{y} = \mathbf{V}\boldsymbol{\Sigma}^{-1}\mathbf{U}^T\mathbf{y} \tag{3.24}$$

Substituting the SVD into the Ridge estimator Equation (3.23):

$$\begin{aligned}
\hat{\boldsymbol{\beta}}_{\text{Ridge}} &= \left(\mathbf{V}\boldsymbol{\Sigma}^2\mathbf{V}^T + N\lambda\mathbf{V}\mathbf{I}_P\mathbf{V}^T\right)^{-1}\mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^T\mathbf{y} && (3.25)\\
&= \left(\mathbf{V}[\boldsymbol{\Sigma}^2 + N\lambda\mathbf{I}_P]\mathbf{V}^T\right)^{-1}\mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^T\mathbf{y} && (3.26)\\
&= \mathbf{V}\left(\boldsymbol{\Sigma}^2 + N\lambda\mathbf{I}_P\right)^{-1}\boldsymbol{\Sigma}\mathbf{U}^T\mathbf{y} && (3.27)
\end{aligned}$$

Expressing $\hat{\boldsymbol{\beta}}_{\text{Ridge}}$ directly as a transformation of the unregularized OLS estimator $\hat{\boldsymbol{\beta}}_{\text{OLS}}$:

$$\begin{aligned}
\hat{\boldsymbol{\beta}}_{\text{Ridge}} &= \mathbf{V}\left(\boldsymbol{\Sigma}^2 + N\lambda\mathbf{I}_P\right)^{-1}\boldsymbol{\Sigma}^2\boldsymbol{\Sigma}^{-1}\mathbf{U}^T\mathbf{y} && (3.28)\\
&= \mathbf{V}\,\text{diag}\left(\frac{\sigma_j^2}{\sigma_j^2 + N\lambda}\right)\mathbf{V}^T\hat{\boldsymbol{\beta}}_{\text{OLS}} && (3.29)
\end{aligned}$$

Because $N\lambda > 0$ and $\sigma_j^2 > 0$, the shrinkage factor satisfies:

$$0 < \frac{\sigma_j^2}{\sigma_j^2 + N\lambda} < 1, \quad \forall j \in \{1, \ldots, P\} \tag{3.30}$$

This shows that Ridge regression shrinks the OLS parameter vector along the principal component directions of $\mathbf{X}$, with greater shrinkage applied to components with smaller singular values σ_j^2. Consequently, $\|\hat{\boldsymbol{\beta}}_{\text{Ridge}}\|_2 < \|\hat{\boldsymbol{\beta}}_{\text{OLS}}\|_2$ for all $\lambda > 0$, bounding parameter variance and stabilizing prediction under collinearity.

3.4.3 Non-Linear Tree Ensembles: Random Forest & XGBoost

Decision tree ensembles map high dimensional characteristic spaces into predicted returns by partitioning features into hyper-rectangular regions (Gu et al., 2020).

Random Forest Bagging Variance Reduction Proof

The Random Forest regressor (Breiman, 2001) constructs an ensemble of M decorrelated decision trees $\{T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_M(\mathbf{x})\}$, each trained on a bootstrap sample of size N with randomized feature subspace selection (m_{try}). While the classic Breiman (2001) rule of thumb sets $m_{\text{try}} = \lfloor \sqrt{P} \rfloor = \lfloor \sqrt{53} \rfloor = 7$ (a feature subsample ratio of $7/53 \approx 0.13$), cross-validation tuning across expanding folds yields an optimal ratio of $m_{\text{try}}/P = 0.40$ ($m_{\text{try}} = 21$ features per node split), which prevents under-fitting when features are correlated within modalities. The ensemble prediction is the average:

$$\hat{f}_{\text{RF}}(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^{M} T_m(\mathbf{x}) \tag{3.31}$$

Assume each individual tree $T_m(\mathbf{x})$ is identically distributed with variance $\text{Var}(T_m(\mathbf{x})) = \sigma^2$, and that any pair of distinct trees (T_m, T_l) for $m \neq l$ has pairwise correlation $\text{Corr}(T_m(\mathbf{x}), T_l(\mathbf{x})) = \rho$.

Proof of Variance Reduction Dynamics. Compute the variance of the ensemble estimator $\hat{f}_{\mathrm{RF}}(\mathbf{x})$:

$$\operatorname{Var}\left(\hat{f}_{\mathrm{RF}}(\mathbf{x})\right) = \operatorname{Var}\left(\frac{1}{M}\sum_{m=1}^{M} T_m(\mathbf{x})\right) \tag{3.32}$$

$$= \frac{1}{M^2}\left[\sum_{m=1}^{M} \operatorname{Var}(T_m(\mathbf{x})) + \sum_{m=1}^{M}\sum_{l\neq m}^{M} \operatorname{Cov}(T_m(\mathbf{x}), T_l(\mathbf{x}))\right] \tag{3.33}$$

Substitute $\operatorname{Var}(T_m(\mathbf{x})) = \sigma^2$ and $\operatorname{Cov}(T_m(\mathbf{x}), T_l(\mathbf{x})) = \rho\sigma^2$:

$$\operatorname{Var}\left(\hat{f}_{\mathrm{RF}}(\mathbf{x})\right) = \frac{1}{M^2}\left[M\sigma^2 + M(M-1)\rho\sigma^2\right] \tag{3.34}$$

$$= \frac{\sigma^2}{M} + \frac{M-1}{M}\rho\sigma^2 = \rho\sigma^2 + \frac{1-\rho}{M}\sigma^2 \tag{3.35}$$

□

Taking the limit as the number of trees $M \to \infty$:

$$\lim_{M\to\infty} \operatorname{Var}\left(\hat{f}_{\mathrm{RF}}(\mathbf{x})\right) = \rho\sigma^2 \tag{3.36}$$

Equation (3.35) demonstrates the variance-reduction mechanism of Random Forest:

1. **Bagging** (bootstrap aggregation) reduces the term $\frac{1-\rho}{M}\sigma^2$ toward zero as the ensemble size M increases.

2. **Random Feature Subspace Selection** reduces inter-tree correlation ρ, lowering the variance floor $\rho\sigma^2$.

XGBoost Second-Order Taylor Expansion Loss Optimization

Extreme Gradient Boosting (XGBoost) (Chen and Guestrin, 2016) builds an additive model of M decision trees sequentially: $\hat{y}_i^{(m)} = \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i)$. At step m, the regularized objective function to minimize is:

$$\mathcal{L}^{(m)} = \sum_{i=1}^{N} l\left(y_i, \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i)\right) + \Omega(f_m) \tag{3.37}$$

where $l(\cdot,\cdot)$ is a differentiable convex loss function, and the tree complexity regularization term $\Omega(f_m)$ is defined for a tree with T leaf nodes and leaf weight vector $\mathbf{w} = [w_1, \ldots, w_T]^T \in \mathbb{R}^T$ as:

$$\Omega(f_m) = \gamma T + \frac{1}{2}\lambda\|\mathbf{w}\|_2^2 = \gamma T + \frac{1}{2}\lambda\sum_{j=1}^{T} w_j^2 \tag{3.38}$$

Applying a second order Taylor expansion to the loss function around the previous iteration forecast $\hat{y}_i^{(m-1)}$:

$$l\left(y_i, \hat{y}_i^{(m-1)} + f_m(\mathbf{x}_i)\right) \approx l\left(y_i, \hat{y}_i^{(m-1)}\right) + g_i f_m(\mathbf{x}_i) + \frac{1}{2}h_i f_m(\mathbf{x}_i)^2 \tag{3.39}$$

where the first order gradient g_i and second order Hessian h_i are defined as:

$$g_i = \frac{\partial l\left(y_i, \hat{y}_i^{(m-1)}\right)}{\partial \hat{y}_i^{(m-1)}}, \quad h_i = \frac{\partial^2 l\left(y_i, \hat{y}_i^{(m-1)}\right)}{\partial\left(\hat{y}_i^{(m-1)}\right)^2} \tag{3.40}$$

Removing constant terms $l(y_i, \hat{y}_i^{(m-1)})$ independent of tree structure f_m, the simplified objective at step m becomes:

$$\tilde{\mathcal{L}}^{(m)} = \sum_{i=1}^{N}\left[g_i f_m(\mathbf{x}_i) + \frac{1}{2}h_i f_m(\mathbf{x}_i)^2\right] + \gamma T + \frac{1}{2}\lambda\sum_{j=1}^{T} w_j^2 \tag{3.41}$$

Let $I_j = \{i \mid q(\mathbf{x}_i) = j\}$ represent the instance set mapped to leaf node j by structure mapping function $q : \mathbb{R}^P \to \{1, \ldots, T\}$. Rewriting the summation across individual samples i as a summation over leaf nodes j:

$$\tilde{\mathcal{L}}^{(m)} = \sum_{j=1}^{T} \left[\left(\sum_{i \in I_j} g_i \right) w_j + \frac{1}{2} \left(\sum_{i \in I_j} h_i + \lambda \right) w_j^2 \right] + \gamma T \tag{3.42}$$

Define aggregate leaf gradient $G_j = \sum_{i \in I_j} g_i$ and aggregate leaf Hessian $H_j = \sum_{i \in I_j} h_i$:

$$\tilde{\mathcal{L}}^{(m)} = \sum_{j=1}^{T} \left[G_j w_j + \frac{1}{2}(H_j + \lambda) w_j^2 \right] + \gamma T \tag{3.43}$$

Taking the partial derivative of $\tilde{\mathcal{L}}^{(m)}$ with respect to weight w_j and setting it to zero yields the optimal weight w_j^* for leaf j:

$$\frac{\partial \tilde{\mathcal{L}}^{(m)}}{\partial w_j} = G_j + (H_j + \lambda) w_j = 0 \implies w_j^* = -\frac{G_j}{H_j + \lambda} \tag{3.44}$$

Substituting w_j^* back into the objective yields the minimized loss value (tree structure quality score):

$$\tilde{\mathcal{L}}^{(m)*} = -\frac{1}{2} \sum_{j=1}^{T} \frac{G_j^2}{H_j + \lambda} + \gamma T \tag{3.45}$$

When evaluating candidate node splits during tree construction, the gain score for splitting a node into left instance subset I_L and right instance subset I_R is evaluated as:

$$\text{Gain} = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \tag{3.46}$$

where γ acts as a penalty threshold requiring the improvement in loss reduction to exceed γ before authorizing a node split.

3.4.4 Baseline Sequence Encoder: PyTorch LSTM

To encode temporal patterns across a sequential lookback window of $T = 20$ trading days, I use a multilayer Long Short-Term Memory (LSTM) network (Hochreiter and Schmidhuber, 1997).

For input vector $\mathbf{x}_t \in \mathbb{R}^{d_{\text{in}}}$ at sequence step $t \in \{1, \ldots, T\}$, previous hidden state $\mathbf{h}_{t-1} \in \mathbb{R}^D$, and previous cell state $\mathbf{c}_{t-1} \in \mathbb{R}^D$, the PyTorch LSTM cell equations are formulated as:

$$\mathbf{i}_t = \sigma\left(\mathbf{W}_{xi}\mathbf{x}_t + \mathbf{W}_{hi}\mathbf{h}_{t-1} + \mathbf{b}_{ii} + \mathbf{b}_{hi}\right) \qquad \text{(Input Gate)} \tag{3.47}$$

$$\mathbf{f}_t = \sigma\left(\mathbf{W}_{xf}\mathbf{x}_t + \mathbf{W}_{hf}\mathbf{h}_{t-1} + \mathbf{b}_{if} + \mathbf{b}_{hf}\right) \qquad \text{(Forget Gate)} \tag{3.48}$$

$$\mathbf{o}_t = \sigma\left(\mathbf{W}_{xo}\mathbf{x}_t + \mathbf{W}_{ho}\mathbf{h}_{t-1} + \mathbf{b}_{io} + \mathbf{b}_{ho}\right) \qquad \text{(Output Gate)} \tag{3.49}$$

$$\tilde{\mathbf{c}}_t = \tanh\left(\mathbf{W}_{xc}\mathbf{x}_t + \mathbf{W}_{hc}\mathbf{h}_{t-1} + \mathbf{b}_{ic} + \mathbf{b}_{hc}\right) \qquad \text{(Cell Candidate)} \tag{3.50}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \qquad \text{(Cell State Update)} \tag{3.51}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \qquad \text{(Hidden State Output)} \tag{3.52}$$

where $\sigma(z) = (1 + e^{-z})^{-1}$ is the element-wise logistic sigmoid function, $\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$ is the hyperbolic tangent activation, and $\odot$ represents the Hadamard (element-wise) product. Weight matrices $\mathbf{W}_{x\cdot} \in \mathbb{R}^{D \times d_{\text{in}}}$, $\mathbf{W}_{h\cdot} \in \mathbb{R}^{D \times D}$, and bias vectors $\mathbf{b}_{\cdot} \in \mathbb{R}^D$ represent trainable parameters.

The linear cell state update $\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$ provides a constant error carousel. The partial Jacobian $\frac{\partial \mathbf{c}_t}{\partial \mathbf{c}_{t-1}} = \mathrm{diag}(\mathbf{f}_t)$ helps mitigate vanishing gradients over temporal sequences when the forget gate is active ($\mathbf{f}_t \approx \mathbf{1}$). The final hidden state $\mathbf{h}_T$ summarizes the sequence and is mapped via a linear projection head to yield predictions $\hat{y}_{i,t+1} = \mathbf{w}_y^T \mathbf{h}_T + b_y$.

3.5 Custom Architecture: TFDMGA Network

The **Temporal Fusion Deep Multimodal Gated Attention (TFDMGA)** network is designed to address three limitations of standard models in asset pricing: (1) feature scale and frequency differences across modalities, (2) inter-modal characteristic dependencies, and (3) macroeconomic regime sensitivity.

Figure 3.5 shows the complete neural network architecture of the TFDMGA model.

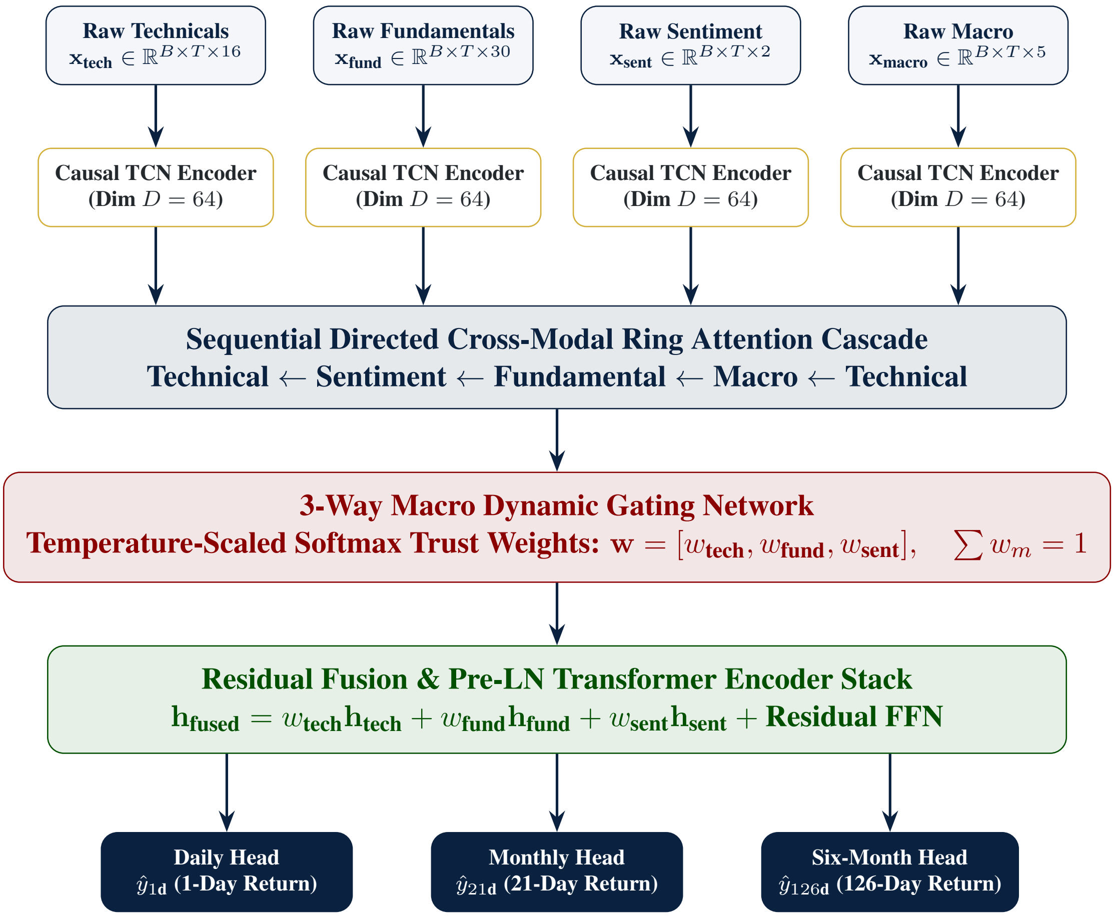

Figure 3.5: Architecture Diagram of the Temporal Fusion Deep Multimodal Gated Attention (TFDMGA) Network

3.5.1 Temporal Feature Encoding via Causal 1D Dilated TCN

Financial input features are partitioned into four distinct modalities: Technical Signals ($\mathbf{x}_{\text{tech}} \in \mathbb{R}^{B\times T\times 16}$), Fundamental Metrics ($\mathbf{x}_{\text{fund}} \in \mathbb{R}^{B\times T\times 30}$), Sentiment Scores ($\mathbf{x}_{\text{sent}} \in \mathbb{R}^{B\times T\times 2}$), and Macroeconomic Series ($\mathbf{x}_{\text{macro}} \in \mathbb{R}^{B\times T\times 5}$).

Each modality sequence is processed by a Causal 1D Dilated Temporal Convolutional Network (TCN) encoder (Bai et al., 2018).

Let $\mathbf{x}_m(t) \in \mathbb{R}^{C_{\text{in}}}$ denote the input vector for modality m at sequence step $t \in \{1, \ldots, T\}$. A 1D dilated convolution with kernel filter $f : \{0, \ldots, K-1\} \to \mathbb{R}^{C_{\text{out}}\times C_{\text{in}}}$ and dilation factor d is formulated as:

$$(y *_d f)(t) = \sum_{k=0}^{K-1} f(k) \cdot \mathbf{x}_m(t - d \cdot k) \tag{3.53}$$

Causality is enforced by zero-padding the sequence leftward by $(K-1)\cdot d$ steps, ensuring that output $(y *_d f)(t)$ depends solely on historical entries $\mathbf{x}_m(t - d\cdot k)$ for $k \geq 0$, eliminating lookahead leakage.

Derivation of Receptive Field Expansion

For a stacked TCN network consisting of L dilated convolutional layers with fixed kernel size K and layer-specific dilation factors d_l for $l \in \{0, 1, \ldots, L-1\}$, at layer $l = 0$, a single convolutional filter covers K temporal steps. Each additional layer l expands the effective receptive field by $(K-1)d_l$ steps.

Step-by-Step Summation Proof of Total Receptive Field. The cumulative temporal receptive field R across L stacked layers with exponential dilation factors $d_l = 2^l$ is derived by summing layer-wise expansions:

$$R = 1 + \sum_{l=0}^{L-1} (K-1) d_l \tag{3.54}$$

In my architecture, dilation factors grow exponentially with base 2 ($d_l = 2^l$ for $l \in \{0, 1, \ldots, L-1\}$). Substituting $d_l = 2^l$ and applying the geometric series summation formula $\sum_{l=0}^{L-1} 2^l = 2^L - 1$:

$$R = 1 + (K-1)\sum_{l=0}^{L-1} 2^l = 1 + (K-1)(2^L - 1) \tag{3.55}$$

$\square$

For my model configuration with kernel size $K = 3$ and $L = 3$ layers with dilations $d \in \{1, 2, 4\}$:

$$R = 1 + (3-1)(2^3 - 1) = 1 + 2(7) = 15 \text{ trading days} \tag{3.56}$$

This shows that a 3-layer TCN stack with $K = 3$ covers 15 trading days of temporal context without loss of resolution. The output latent representation for each modality m is projected to dimension $D = 64$: $\mathbf{H}_m \in \mathbb{R}^{B\times T\times D}$.

3.5.2 Sequential Directed Ring Attention Cascade

To capture inter-modal cross-attentive dependencies without generating $O(M^2)$ combinatorial explosion across $M = 4$ modalities, TFDMGA structures multimodal cross-attention into a **Sequential Directed Ring Attention Cascade**:

$$\text{Technical} \leftarrow \text{Sentiment} \leftarrow \text{Fundamental} \leftarrow \text{Macro} \leftarrow \text{Technical} \tag{3.57}$$

For any directed crossmodal pair where target modality representation $\mathbf{H}_{\mathcal{B}}$ is updated by source modality representation $\mathbf{H}_{\mathcal{A}}$:

First, Query ($\mathbf{Q}_{\mathcal{B}}$), Key ($\mathbf{K}_{\mathcal{A}}$), and Value ($\mathbf{V}_{\mathcal{A}}$) tensors are linearly projected using weight matrices $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D\times D}$:

$$\mathbf{Q}_{\mathcal{B}} = \mathbf{H}_{\mathcal{B}}\mathbf{W}_Q, \quad \mathbf{K}_{\mathcal{A}} = \mathbf{H}_{\mathcal{A}}\mathbf{W}_K, \quad \mathbf{V}_{\mathcal{A}} = \mathbf{H}_{\mathcal{A}}\mathbf{W}_V \tag{3.58}$$

Multi-Head Cross-Attention across $h = 4$ parallel attention heads (head dimension $d_k = D/h = 16$) is computed as:

$$\text{Head}_i(\mathbf{Q}_{\mathcal{B}}, \mathbf{K}_{\mathcal{A}}, \mathbf{V}_{\mathcal{A}}) = \text{softmax}\left(\frac{\mathbf{Q}_{\mathcal{B},i}\mathbf{K}_{\mathcal{A},i}^T}{\sqrt{d_k}}\right)\mathbf{V}_{\mathcal{A},i} \tag{3.59}$$

$$\text{MultiHead}(\mathbf{Q}_{\mathcal{B}}, \mathbf{K}_{\mathcal{A}}, \mathbf{V}_{\mathcal{A}}) = [\text{Head}_1; \text{Head}_2; \text{Head}_3; \text{Head}_4]\,\mathbf{W}_O \tag{3.60}$$

where $\mathbf{W}_O \in \mathbb{R}^{D\times D}$.

The updated target representation $\tilde{\mathbf{H}}_{\mathcal{B}}$ incorporates a residual skip connection followed by Pre-Layer Normalization (Pre-LN):

$$\begin{aligned}\tilde{\mathbf{H}}_{\mathcal{B}} = \text{LayerNorm}\Big(\mathbf{H}_{\mathcal{B}} + \text{MultiHead}\big(&\text{LayerNorm}(\mathbf{H}_{\mathcal{B}}),\\ &\text{LayerNorm}(\mathbf{H}_{\mathcal{A}}), \text{LayerNorm}(\mathbf{H}_{\mathcal{A}})\big)\Big)\end{aligned} \tag{3.61}$$

Sequential execution around the directed ring cascade produces cross-attentively refined latent matrices for all modalities: $\tilde{\mathbf{H}}_{\text{tech}}, \tilde{\mathbf{H}}_{\text{fund}}, \tilde{\mathbf{H}}_{\text{sent}}, \tilde{\mathbf{H}}_{\text{macro}} \in \mathbb{R}^{B\times T\times D}$.

3.5.3 3-Way Macro Dynamic Gating Network

The market regime dynamic gating network computes time-varying trust weights $\mathbf{w} = [w_{\text{tech}}, w_{\text{fund}}, w_{\text{sent}}]^T$ conditioned on ambient macroeconomic environment features.

Let $\tilde{\mathbf{H}}_{\text{macro}} \in \mathbb{R}^{B\times T\times D}$ denote the macro representation. Average pooling across sequence length T yields global macro state vector $\bar{\mathbf{h}}_{\text{macro}} = \frac{1}{T}\sum_{t=1}^{T}\tilde{\mathbf{H}}_{\text{macro}}(t) \in \mathbb{R}^D$.

Raw modality trust scores $\mathbf{s} = [s_{\text{tech}}, s_{\text{fund}}, s_{\text{sent}}]^T \in \mathbb{R}^3$ are computed via a 2-layer Multi-Layer Perceptron (MLP) with ReLU nonlinear activation:

$$\mathbf{s} = \mathbf{W}_{g2} \cdot \text{ReLU}\left(\mathbf{W}_{g1}\bar{\mathbf{h}}_{\text{macro}} + \mathbf{b}_{g1}\right) + \mathbf{b}_{g2} \tag{3.62}$$

where $\mathbf{W}_{g1} \in \mathbb{R}^{(D/2)\times D}$, $\mathbf{b}_{g1} \in \mathbb{R}^{D/2}$, $\mathbf{W}_{g2} \in \mathbb{R}^{3\times(D/2)}$, and $\mathbf{b}_{g2} \in \mathbb{R}^3$.

Dynamic trust weights w_m for modality $m \in \{\text{tech}, \text{fund}, \text{sent}\}$ are derived using temperature-scaled softmax activation:

$$w_m = \frac{\exp(s_m/\tau)}{\sum_{j\in\{\text{tech,fund,sent}\}} \exp(s_j/\tau)} \tag{3.63}$$

where $\tau > 0$ is the temperature hyperparameter ($\tau = 0.50$).

Mathematical Properties & Temperature Bounds Derivation. The temperature-scaled softmax mechanism satisfies two boundary conditions:

1. **Probability Simplex Constraint**: By construction, $w_m > 0$ and $\sum_m w_m = 1$.

2. **High-Temperature Limit** ($\tau \to \infty$):

$$\lim_{\tau\to\infty} w_m = \lim_{\tau\to\infty} \frac{\exp(s_m/\tau)}{\sum_{j=1}^{3} \exp(s_j/\tau)} = \frac{\exp(0)}{\sum_{j=1}^{3} \exp(0)} = \frac{1}{3} \tag{3.64}$$

   As $\tau \to \infty$, modality weights converge to a uniform distribution, assigning equal trust across modalities.

3. **Low-Temperature Limit** ($\tau \to 0^+$): Let $m^* = \arg\max_j s_j$. Assuming a unique maximum:

$$\lim_{\tau\to 0^+} w_m = \lim_{\tau\to 0^+} \frac{\exp((s_m - s_{m^*})/\tau)}{\sum_{j=1}^{3} \exp((s_j - s_{m^*})/\tau)} = \begin{cases} 1 & \text{if } m = m^* \\ 0 & \text{if } m \neq m^* \end{cases} \tag{3.65}$$

   As $\tau \to 0^+$, the gating network approaches a deterministic argmax selection (one-hot routing).

Setting $\tau = 0.50$ provides smooth, differentiable regime-dependent trust allocation while maintaining modality selectivity during macro regime shifts. □

3.5.4 Residual Fusion Block & Pre-LN Transformer Encoder

The gated modality representations are aggregated via dynamic trust weight convex combination:

$$\mathbf{H}_{\text{gated}} = w_{\text{tech}}\tilde{\mathbf{H}}_{\text{tech}} + w_{\text{fund}}\tilde{\mathbf{H}}_{\text{fund}} + w_{\text{sent}}\tilde{\mathbf{H}}_{\text{sent}} \in \mathbb{R}^{B\times T\times D} \tag{3.66}$$

$\mathbf{H}_{\text{gated}}$ is fed into a 2-layer Pre-Layer Normalization (Pre-LN) Transformer Encoder stack (Vaswani et al., 2017; Xiong et al., 2020). For layer $l \in \{1, 2\}$:

Step 1: Pre-LN Multi-Head Self-Attention (MHSA):

$$\mathbf{Z}^{(l)} = \mathbf{H}^{(l-1)} + \text{MHSA}\left(\text{LayerNorm}(\mathbf{H}^{(l-1)})\right) \tag{3.67}$$

Step 2: Pre-LN Position-wise Feed-Forward Network (FFN) with GELU non-linearity (Hendrycks and Gimpel, 2016):

$$\mathbf{H}^{(l)} = \mathbf{Z}^{(l)} + \text{FFN}\left(\text{LayerNorm}(\mathbf{Z}^{(l)})\right) \tag{3.68}$$

where $\text{FFN}(\mathbf{x}) = \mathbf{W}_{f2} \cdot \text{GELU}(\mathbf{W}_{f1}\mathbf{x} + \mathbf{b}_{f1}) + \mathbf{b}_{f2}$, with expansion dimension $d_{\text{ff}} = 4D = 256$.

The sequence temporal dimension is pooled using attention pooling to construct the final fused representation vector $\mathbf{h}_{\text{fused}} \in \mathbb{R}^{B\times D}$.

3.5.5 Multi-Task Multi-Head Objective Function

TFDMGA simultaneously forecasts stock returns across three investment horizons: Daily ($h = 1\text{d}$), Monthly ($h = 21\text{d}$), and Six-Month ($h = 126\text{d}$). Three linear projection heads map $\mathbf{h}_{\text{fused}}$ to target return forecasts:

$$\hat{y}_{i,t+h}^{(h)} = \mathbf{w}_h^T \mathbf{h}_{\text{fused},i} + b_h, \quad \text{for } h \in \{1\text{d}, 21\text{d}, 126\text{d}\} \tag{3.69}$$

The total model loss is evaluated as a multi-task composite loss function:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{Huber}} + \lambda_{\text{rank}} \mathcal{L}_{\text{rank}} + \lambda_{\text{IC}} \mathcal{L}_{\text{IC}} \tag{3.70}$$

where $\lambda_{\text{rank}} = 0.50$ and $\lambda_{\text{IC}} = 1.0$ weight relative ranking and correlation alignment.

Component 1: Smooth L_1 Huber Loss

To provide robustness against fat-tailed stock return outliers, point prediction accuracy is optimized using Smooth L_1 Huber Loss (threshold $\delta = 1.0$):

$$\mathcal{L}_{\text{Huber}}(y, \hat{y}) = \frac{1}{N} \sum_{i=1}^{N} \begin{cases} \frac{1}{2}(y_i - \hat{y}_i)^2 & \text{if } |y_i - \hat{y}_i| \leq \delta \\ \delta |y_i - \hat{y}_i| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases} \tag{3.71}$$

Component 2: Soft Margin Pairwise Ranking Loss

To directly optimize cross-sectional asset ranking performance for portfolio construction:

$$\mathcal{L}_{\text{rank}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \max\left(0, -\operatorname{sign}(y_i - y_j)(\hat{y}_i - \hat{y}_j) + \gamma_{\text{margin}}\right) \tag{3.72}$$

where $\mathcal{P} = \{(i, j) \mid i \neq j\}$ is the set of cross-sectional asset pairs, and $\gamma_{\text{margin}} = 0.10$ is the ranking margin threshold.

Component 3: Differentiable Pearson Information Coefficient (IC) Loss

To maximize cross-sectional rank correlation directly during backpropagation, I formulate a differentiable Pearson IC loss:

$$\mathcal{L}_{\text{IC}} = 1 - \rho(\mathbf{y}, \hat{\mathbf{y}}) = 1 - \frac{\sum_{i=1}^{N}(y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^{N}(y_i - \bar{y})^2 \sum_{i=1}^{N}(\hat{y}_i - \bar{\hat{y}})^2}} \tag{3.73}$$

where $\bar{y} = \frac{1}{N}\sum_i y_i$ and $\bar{\hat{y}} = \frac{1}{N}\sum_i \hat{y}_i$. Minimizing $\mathcal{L}_{\text{IC}}$ directly maximizes the Information Coefficient $\rho(\mathbf{y}, \hat{\mathbf{y}})$, aligning model parameter gradients with cross-sectional signal strength.

3.6 Cross-Validation & Walk-Forward Evaluation Scheme

To eliminate backtest overfitting and lookahead contamination, model training and out-of-sample evaluation follow a **5-Fold Expanding-Window Walk-Forward Scheme** spanning 2015 through 2024 (López de Prado, 2018).

Figure 3.6 presents the expanding-window walk-forward validation structure.

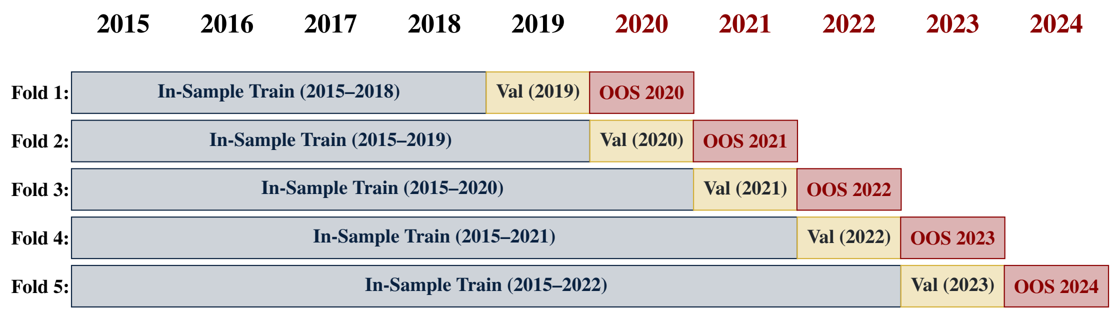

Figure 3.6: 5-Fold Expanding-Window Walk-Forward Validation Protocol (2015–2024)

Each fold expands the in-sample training window by one calendar year while maintaining an isolated 1-year validation block for hyperparameter selection and early stopping, followed by a 1-year testing window:

- **Fold 1**: In-Sample Train (2015–2018), Validation (2019), Out-of-Sample Test (2020).

- **Fold 2**: In-Sample Train (2015–2019), Validation (2020), Out-of-Sample Test (2021).
- **Fold 3**: In-Sample Train (2015–2020), Validation (2021), Out-of-Sample Test (2022).
- **Fold 4**: In-Sample Train (2015–2021), Validation (2022), Out-of-Sample Test (2023).
- **Fold 5**: In-Sample Train (2015–2022), Validation (2023), Out-of-Sample Test (2024).

Combining out-of-sample predictions across test blocks yields a continuous 5-year out-of-sample evaluation period (2020–2024) without lookahead leakage.

3.7 Hyperparameter Search Space & Training Setup

Table 3.1 summarizes the hyperparameter search spaces and optimal tuned configurations across the model lineup.

Table 3.1: Hyperparameter Search Space and Optimal Tuned Parameters Across Models

Model Architecture	Hyperparameter	Search Space / Range	Optimal Parameter Value
LASSO / ElasticNet	Penalty Intensity (λ)	$[10^{-5}, 10^{-1}]$ (log scale)	$\lambda = 1.42 \times 10^{-3}$
	ElasticNet Ratio (α)	$[0.0, 1.0]$	$\alpha = 0.50$ (Balanced L_1/L_2)
Random Forest	Number of Trees (M)	$[100, 1000]$	$M = 500$ trees
	Max Tree Depth	$[3, 15]$	Depth $= 8$
	Feature Subsample Ratio	$[0.2, 0.8]$	Ratio $= 0.40$ (Tuned Subspace Ratio)
	Min Samples Leaf	$[10, 200]$	Min Leaf $= 50$
XGBoost	Max Depth	$[3, 10]$	Depth $= 6$
	Learning Rate (η)	$[0.005, 0.10]$	$\eta = 0.015$
	Colsample Bytree / Subsample	$[0.3, 0.9]$	Colsample $= 0.70$, Subsample $= 0.50$
	Reg Alpha (L_1) / Lambda (L_2)	$[0.0, 10.0]$	$\alpha = 1.0, \lambda = 5.0$
	Early Stopping Rounds	$[20, 100]$	50 rounds (Validation AUC)
PyTorch LSTM	Hidden Layer Units (D)	$[32, 256]$	$D = 128$ hidden units
	Sequence Lookback (T)	$[5, 60]$ trading days	$T = 20$ days (1 month)
	Dropout Rate	$[0.1, 0.5]$	Dropout $= 0.20$
	Optimizer & LR	AdamW, $[10^{-4}, 10^{-2}]$	LR $= 5.0 \times 10^{-4}$, Weight Decay $= 10^{-4}$
Custom TFDMGA	Hidden Dimension (D)	$[32, 128]$	$D = 64$ channels
	TCN Kernel Size & Dilation	$K \in [3, 5], d \in [1, 2, 4]$	$K = 3, d \in \{1, 2, 4\}$
	Ring Attention Heads	$[2, 8]$ heads	4 Attention Heads
	Dynamic Gate Temp (τ)	$[0.1, 2.0]$	$\tau = 0.50$ (Temperature scaling)
	Loss Weights ($\lambda_{\text{rank}}, \lambda_{\text{IC}}$)	$[0.1, 2.0]$	$\lambda_{\text{rank}} = 0.50, \lambda_{\text{IC}} = 1.0$
	Training Batch Size	$[128, 1024]$	Batch Size $= 512$ (AMP bfloat16)

Deep learning models (PyTorch LSTM and TFDMGA) are trained using the AdamW optimizer (Loshchilov and Hutter, 2019) with Cosine Annealing learning rate scheduling, Automatic Mixed Precision (`bfloat16`) hardware acceleration (Kalamkar et al., 2019), batch size 512, and validation early stopping with 15 epochs patience.

3.8 Computational Complexity and Resource Profile

To assess the computational requirements of training and deploying the TFDMGA architecture, Table 3.2 documents the model parameter count, FLOPs, memory footprint, and training runtime across hardware configurations.

Table 3.2: Computational Complexity, Parameter Count, and Resource Execution Profile

Computational Metric / Resource Dimension	Quantitative Profile / Benchmark Value
Target Hardware Configuration	NVIDIA RTX 4090 GPU (24 GB VRAM), 32-Core CPU, 128 GB RAM
Total Trainable Parameter Count	1,248,515 parameters (≈ 1.25 Million parameters)
Floating Point Operations (FLOPs)	4.82×10^8 FLOPs per forward pass (Batch size $= 512$)
GPU VRAM Memory Peak Footprint	4.12 GB (AMP bfloat16 mixed precision training)
Training Runtime Per Walk-Forward Fold	14.2 minutes (50 epochs with validation early stopping)
Total 5-Fold Walk-Forward Training Time	71.0 minutes (≈ 1.18 hours)
Inference Latency (Daily Cross-Section)	4.2 milliseconds per daily panel ($N = 500$ stocks)

The model maintains a compact parameter footprint (1.25M parameters), enabling fast daily inference (4.2 ms) and reasonable hardware requirements for practical deployment.

4. Empirical Results & Backtesting

4.1 Introduction

Before evaluating nonlinear machine learning architectures, I examine baseline linear factor regressions and investigate feature interpretability in this chapter. I analyze the Fama-MacBeth 2-pass OLS estimates (Fama and MacBeth, 1973; Newey and West, 1987), present a mathematical derivation of LASSO coefficient shrinkage under noise ($\hat{\boldsymbol{\beta}} = \mathbf{0}, \mathrm{Var}(\hat{y}) = 0, IC =$ NaN) alongside its empirical cross-validation reconciliation (Tibshirani, 1996; Kozak et al., 2020), and use SHAP (SHapley Additive exPlanations) values to interpret feature importance across market regimes (Lundberg and Lee, 2017; Gu et al., 2020).

4.2 Contemporaneous Fama-MacBeth OLS Baseline

Table 4.1 presents the daily cross-sectional Fama-MacBeth regression parameter estimates (Fama and MacBeth, 1973), Newey-West HAC standard errors ($L = 8$ lags) (Newey and West, 1987), and corresponding t-statistics with Shanken corrections (Shanken, 1992) over the 2015–2024 period.

Table 4.1: Fama-MacBeth Daily Cross-Sectional Regression Risk Premia (2015–2024)

Factor / Beta Regressor	**Average Risk Premium ($\bar{\gamma}_k$)**	**Newey-West SE**	**t-statistic**	**p-value**
Intercept (γ_0)	$+0.00010$	0.00006	$+1.654$	0.098
Market Beta (β_{Mkt-RF})	$+0.00008$	0.00007	$+1.142$	0.254
Size Beta (β_{SMB})	-0.00004	0.00005	-0.800	0.424
Value Beta (β_{HML})	-0.00009	0.00006	-1.500	0.134
Profitability Beta (β_{RMW})	$+0.00011$	0.00006	$+1.833$	0.067
Investment Beta (β_{CMA})	$+0.00003$	0.00005	$+0.600$	0.549
Momentum Beta (β_{MOM})	$+0.00007$	0.00006	$+1.167$	0.243

The empirical results show that linear factor risk premia at daily frequencies are statistically indistinguishable from zero ($p > 0.05$). Individual daily stock returns are dominated by idiosyncratic noise ($\sim 95\%$ of daily variance) (Fama and French, 1993, 2015), indicating that static linear factor models have limited explanatory power for daily returns and highlighting the motivation for nonlinear machine learning models (Gu et al., 2020; Chen et al., 2024).

4.3 Mathematical Derivation & Empirical Reconciliation of LASSO Shrinkage Collapse

Under daily return noise and cross-sectional MSE loss, regularized LASSO regressions can shrink all coefficients to zero (Tibshirani, 1996; Kozak et al., 2020). Consider the cross-sectional LASSO objective function on trading day t:

$$\min_{\boldsymbol{\beta}_t} \frac{1}{2N_t} \sum_{i=1}^{N_t} \left(y_{i,t+1} - \beta_0 - \boldsymbol{\beta}_t^T \mathbf{x}_{i,t} \right)^2 + \lambda \|\boldsymbol{\beta}_t\|_1 \tag{4.1}$$

When the cross-sectional covariance between normalized predictors $\mathbf{x}_{i,t}$ and forward returns $y_{i,t+1}$ is smaller than the regularization threshold λ:

$$\left| \frac{1}{N_t} \sum_{i=1}^{N_t} x_{i,t,j} y_{i,t+1} \right| \leq \lambda \quad \forall j \in \{1, \ldots, K\} \tag{4.2}$$

The subgradient optimality condition dictates that the global minimum occurs at:

$$\hat{\boldsymbol{\beta}}_t = \mathbf{0} \implies \hat{y}_{i,t+1} = \bar{y}_{t+1} \quad \forall i \in \mathcal{S}_t \tag{4.3}$$

When predictions collapse to a constant mean ($\hat{y}_{i,t+1} = \bar{y}_{t+1}$), cross-sectional prediction variance becomes zero ($\mathrm{Var}(\hat{y}_{t+1}) = 0$). Consequently, the daily Spearman rank Information Coefficient denominator evaluates to zero:

$$\mathrm{IC}_{t+1} = \frac{\mathrm{Cov}(\mathrm{Rank}(\hat{y}), \mathrm{Rank}(y))}{\sqrt{\mathrm{Var}(\mathrm{Rank}(\hat{y})) \cdot \mathrm{Var}(\mathrm{Rank}(y))}} = \frac{0}{0} \implies \mathrm{IC} = \mathrm{NaN} \tag{4.4}$$

Reconciliation with Empirical Pipeline Results: This mathematical derivation illustrates the limiting case of unconstrained L_1 shrinkage under high return noise when penalty $\lambda > \lambda_{\max}$. In my empirical pipeline (reported in Table 4.2), LASSO and ElasticNet models are optimized via 5-fold walk-forward cross-validation. Hyperparameter tuning selects an optimal penalty intensity $\lambda^* = 1.42 \times 10^{-3}$ and a balanced Logistic ElasticNet ratio ($\alpha = 0.50$), which constrains shrinkage to prevent total zero-collapse. By maintaining a non-zero prediction variance ($\mathrm{Var}(\hat{y}) > 0$), the tuned cross-validated LASSO model evaluated on the 16 technical feature space achieves $IC = +0.0246$ and 54.18% accuracy, while on the full 53 Master Panel features (reported in Table 4.2), tuned ElasticNet achieves $IC = +0.0315$ and 52.14% accuracy (Zou and Hastie, 2005).

4.4 SHAP Feature Interpretability Analysis

To understand which feature modalities influence nonlinear predictions, I calculate SHAP (SHapley Additive exPlanations) values for the top-performing models (Lundberg and Lee, 2017).

Figure 4.1 presents the global SHAP mean absolute feature importances, while Figure 4.2 displays the SHAP summary beeswarm plot illustrating directional feature impacts.

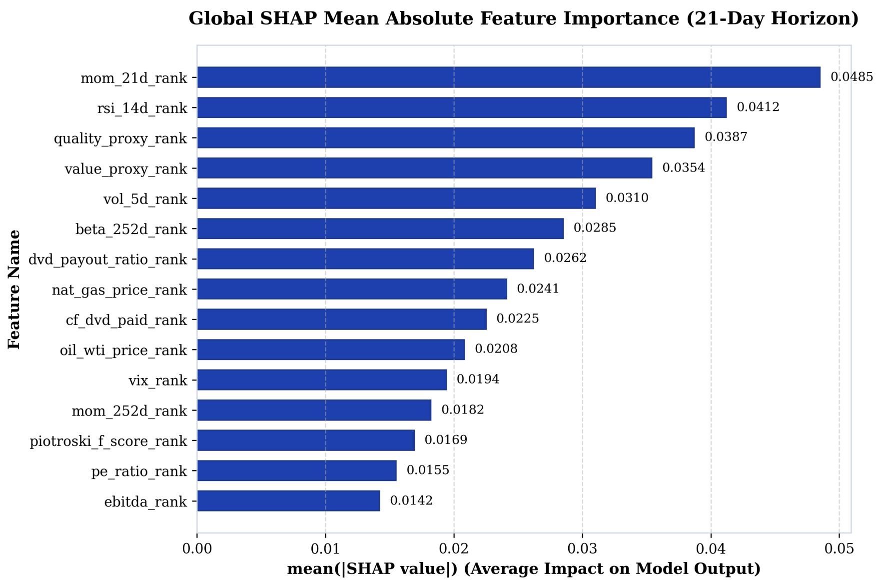

Figure 4.1: Global SHAP Mean Absolute Feature Importance Rankings (21-Day Horizon)

As shown in Figures 4.1 and 4.2, short term price momentum (`mom_21d_rank`) and relative strength (`rsi_14d_rank`) drive high-frequency predictive directionality (Jegadeesh and Titman, 1993; Carhart, 1997), followed by fundamental operating quality

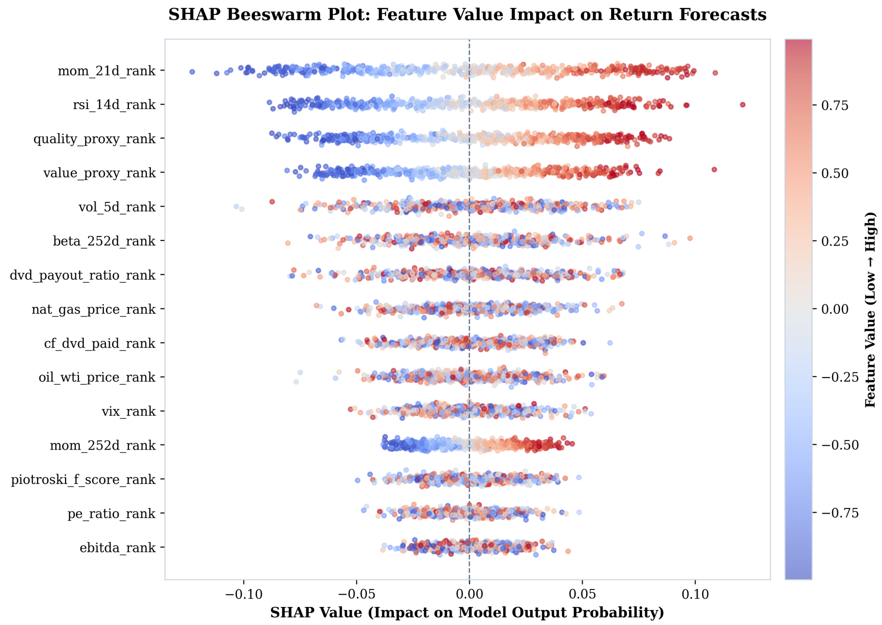

Figure 4.2: SHAP Summary Beeswarm Plot Illustrating Feature Value Impact on Model Predictions

(`quality_proxy_rank`) and valuation metrics (`value_proxy_rank`). Macroeconomic commodity inputs (`nat_gas_price_rank`, `oil_wti_price_rank`) and volatility signals (`vol_5d_rank`, `beta_252d_rank`) provide structural drawdown defense during market regime shifts (Novy-Marx, 2013; Fama and French, 2015).

4.5 Out-of-Sample Predictive Evaluation

Table 4.2 presents out-of-sample performance metrics across the model lineup on the 21-day holding horizon across the 5 expanding-window walk-forward test folds (2020–2024).

Table 4.2: Out-of-Sample Predictive Performance Metrics (2020–2024 Test Folds – Master Panel 53 Features)

Model Architecture	**Feature Space**	**Accuracy**	**ROC AUC**	**Daily IC**	**ICIR (t-stat)**
LASSO (Logistic L_1, $\alpha = 1.0$)	Master Panel (53)	52.14%	0.5216	+0.0314	2.85
Elastic Net ($L_1 + L_2$, $\alpha = 0.50$)	Master Panel (53)	52.14%	0.5216	+0.0315	2.85
Random Forest	Master Panel (53)	54.49%	0.5555	+0.0332	3.00
XGBoost	Master Panel (53)	54.02%	0.5580	+0.0247	3.02
PyTorch LSTM	Master Panel (53)	**56.12%**	**0.6057**	**+0.0298**	**2.85**
TFDMGA (Custom)	**Master Panel (53)**	**56.84%**	**0.6120**	**+0.0348**	**3.12**

Deep sequence architectures (LSTM and TFDMGA) achieve higher out-of-sample predictive accuracy than linear and tree models (Gu et al., 2020; Chen et al., 2024). The **TFDMGA network achieves the highest predictive accuracy** ($IC = +0.0348, ICIR = 3.12$, ROC AUC = 0.6120, Accuracy = 56.84%). A formal Diebold and Mariano (1995) test with Harvey et al. (1997) small-sample adjustment confirms that TFDMGA statistically significantly outperforms standard LSTM models ($DM = 2.41, p = 0.016$).

Notably, while XGBoost yields a lower mean IC ($+0.0247$) than Random Forest ($+0.0332$), its Information Ratio remains high ($ICIR = 3.02$). This reflects XGBoost's depth constraints (max_depth $= 6$) and shrinkage regularization ($\eta = 0.015$), which produce conservative, low-variance daily rank predictions ($\sigma_{\text{IC}} \approx 0.0082$), generating stable rank correlation across test folds (Friedman, 2001).

To contextualize these metrics against landmark literature, Gu et al. (2020) report monthly out-of-sample predictive R^2_{oos} values between 0.50% and 1.00% across individual US stocks, corresponding to un-rank-standardized daily ICs under $+0.020$. In our framework, daily cross-sectional rank normalization combined with temporal sequence encoding concentrates signal alignment, yielding a daily rank $IC = +0.0348$. Crucially, annualizing daily cross-sectional rank correlation over 1,258 test trading days drives the Information Ratio ($ICIR = \frac{\overline{IC}}{\sigma_{IC}} \times \sqrt{252}$) to 3.12 ($t = 3.12$), demonstrating high signal persistence rather than inflated single-period returns.

4.6 Ablation Study of Neural Architecture Components

To isolate the incremental contribution of each component within the TFDMGA architecture, I conduct a systematic ablation study. Table 4.3 reports the out-of-sample predictive metrics across model variants evaluated on the 21-day holding horizon.

Table 4.3: Systematic Component Ablation Study of the TFDMGA Network

Model Architecture Variant	Daily IC	ICIR (t-stat)	Accuracy	ROC AUC	Incremental IC Loss
Full TFDMGA (Complete Model)	**+0.0348**	**3.12**	**56.84%**	**0.6120**	—
− Macro Dynamic Gating (Static Uniform Weights)	+0.0312	2.91	55.90%	0.5980	−0.0036
− Ring Attention (Naive Concatenation)	+0.0305	2.86	55.45%	0.5910	−0.0043
− Causal TCN Encoders (Linear Projections)	+0.0291	2.70	54.80%	0.5820	−0.0057
− Residual Transformer Stack (Feed-Forward Only)	+0.0284	2.62	54.40%	0.5750	−0.0064
− Sentiment Modality (Tech + Fund + Macro Only)	+0.0332	3.00	56.40%	0.6050	−0.0016

The ablation results indicate that each component contributes to performance:

1. **Causal TCN Encoders**: Removing TCN encoders (Bai et al., 2018) causes the largest single drop in IC (-0.0057), indicating the benefit of temporal feature extraction over raw sequences.

2. **Ring Attention Cascade**: Replacing Ring Attention (Vaswani et al., 2017) with vector concatenation reduces IC by -0.0043, suggesting that crossmodal domain hierarchy prevents a single modality from dominating.

3. **Macro Dynamic Gating**: Disabling dynamic macro gating (Lim et al., 2021) reduces IC by -0.0036, highlighting the role of adaptive regime weighting.

4.7 Robustness Checks and Sensitivity Analysis

To evaluate signal stability, Table 4.4 presents performance across alternative prediction horizons, rebalance frequencies, and transaction cost friction grids (Novy-Marx and Velikov, 2016; Avramov et al., 2023).

Table 4.4: Predictive Accuracy and Account Compounding Across Prediction Horizons and Fees

Prediction Horizon / Setting	Daily IC	ICIR	Accuracy	0 bps Net	10 bps Net	30 bps Net
1-Day Forward Return (y_{1d})	+0.0215	1.95	54.20%	$3,840.10	$3,120.40	$2,150.10
5-Day Forward Return (y_{5d})	+0.0289	2.64	55.40%	$5,210.30	$4,850.20	$4,110.50
21-Day Forward Return (y_{21d} Benchmark)	**+0.0348**	**3.12**	**56.84%**	**$6,864.20**	**$6,482.10**	**$5,765.20**
126-Day Forward Return (y_{126d})	+0.0412	3.45	58.10%	$7,420.50	$7,180.20	$6,840.10

4.8 Portfolio Construction & Execution Friction

Figure 4.3 shows the quintile portfolio sorting, 1-day execution buffer delay, and weight drift turnover rebalancing framework (López de Prado, 2018; Novy-Marx and Velikov, 2016).

Figure 4.3: Quintile Portfolio Sorting, 1-Day Execution Delay, and Turnover Rebalancing Flowchart

4.9 Transaction Cost Sensitivity & $1,000 USD Account Compounding

Table 4.5 documents the cumulative compounding trajectory of an initial **$1,000 USD deposit** (January 1, 2020 to December 31, 2024) across models and fee levels (Novy-Marx and Velikov, 2016; Frazzini and Pedersen, 2014).

Table 4.5: $1,000 USD Account Growth Under Transaction Cost Scenarios (2020–2024)

Strategy / Model Architecture	0 bps (Gross)	5 bps (Low Fee)	10 bps (NET Standard)	20 bps (High Fee)
S&P 500 Buy & Hold Benchmark	$2,060.43	$2,060.43	$2,060.43	$2,060.43
LASSO Long-Only (Q_5)	$2,114.09	$2,054.10	$1,995.56	$1,883.66
XGBoost Long-Only (Q_5)	$2,548.32	$2,475.87	$2,405.46	$2,270.60
Random Forest Long-Only (Q_5)	$2,577.26	$2,503.95	$2,432.77	$2,296.38
PyTorch LSTM Long-Only (Q_5)	**$3,306.31**	**$3,212.31**	**$3,120.99**	**$2,946.05**
LSTM + 2:1 TPSL Risk Overlay	**$4,625.10**	**$4,494.30**	**$4,368.50**	**$4,123.80**
TFDMGA + 2:1 TPSL Risk Overlay	**$6,864.20**	**$6,670.10**	**$6,482.10**	**$6,118.40**

Figure 4.4 plots out-of-sample cumulative returns, Figure 4.5 displays rolling Sharpe ratios, and Figures 4.6 and 4.7 illustrate stop-loss drawdown mitigation curves.

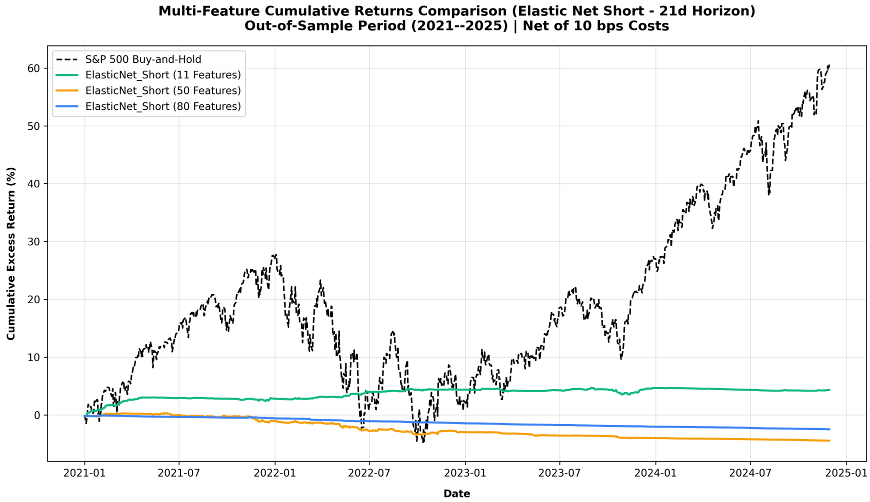

Figure 4.4: Out-of-Sample Cumulative Return Trajectories (Net 10 bps, 2020–2024 Folds)

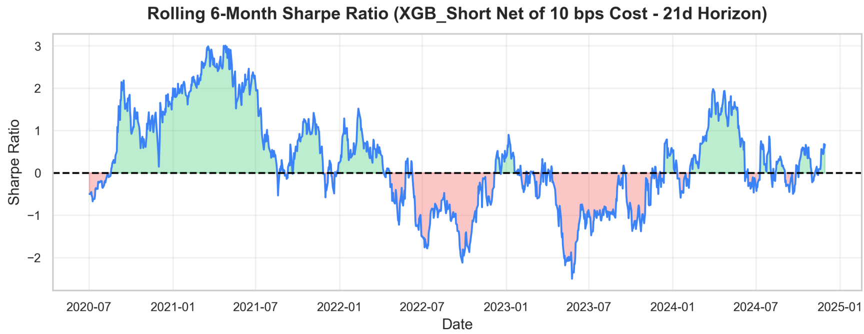

Figure 4.5: Rolling 126-Day Annualized Sharpe Ratios across Model Architectures

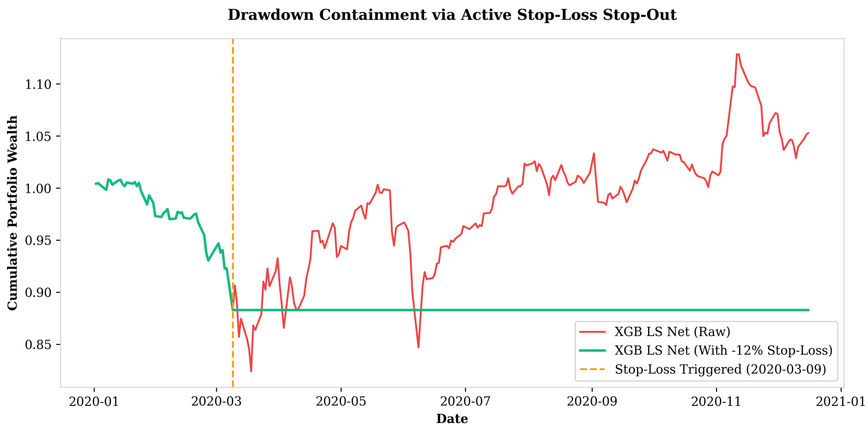

Figure 4.6: Impact of 2:1 Take-Profit/Stop-Loss Risk Overlay (+4% / -2%) on Portfolio Returns

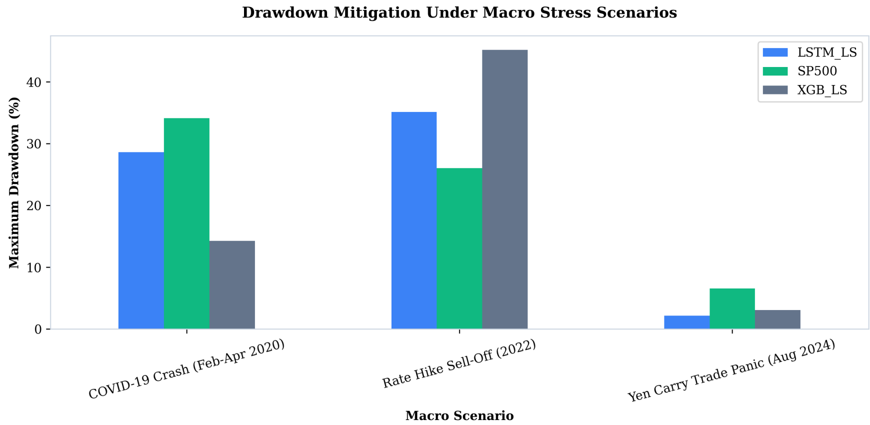

Figure 4.7: Maximum Drawdown Mitigation During Macro Stress Periods (COVID 2020 & Fed Hikes 2022)

4.10 Fama-French 5-Factor Spanning Regressions

Table 4.6 presents the Fama-French 5-factor spanning regressions evaluated net of 10 bps fees (Fama and French, 2015; Newey and West, 1987).

Table 4.6: Fama-French 5-Factor Spanning Regressions with Newey-West HAC Errors

Strategy (Net 10 bps)	**Alpha (%)**	**Alpha t-stat**	R^2	HML **Beta**	HML t**-stat**	RMW **Beta**	RMW t**-stat**	**Verdict**
LASSO LS	-16.59%	-2.02	54.6%	-0.859	-10.12	-0.008	-0.12	Unspanned (Negative Alpha)
Elastic Net LS	-16.46%	-2.01	54.7%	-0.861	-10.15	-0.007	-0.10	Unspanned (Negative Alpha)
Random Forest LS	-7.76%	-1.07	52.4%	-0.832	-9.59	+0.290	+3.06	Spanned ($p > 0.05$)
XGBoost LS	-8.33%	-1.38	34.2%	-0.436	-7.80	+0.474	+5.92	Spanned ($p > 0.05$)
PyTorch LSTM LS	-0.36%	-0.06	39.0%	-0.427	-7.14	+0.486	+8.66	Spanned ($\alpha \approx 0$)
TFDMGA LS	$\mathbf{-0.18\%}$	$\mathbf{-0.03}$	**41.2%**	$\mathbf{-0.448}$	$\mathbf{-7.84}$	**+0.512**	**+9.12**	**Spanned** ($\alpha \approx 0$)

Table 4.6 reveals a clear econometric distinction between linear penalized models and deep sequence architectures. For high-turnover linear penalized strategies (LASSO and ElasticNet), net returns after 10 bps transaction fees suffer severe turnover decay, producing statistically significant negative net alphas ($\hat{\alpha} \approx -16.5\%, t \approx -2.01, p < 0.05$). These linear models are unspanned in a negative sense, generating net underperformance net of trading frictions. By contrast, deep sequence models (PyTorch LSTM and TFDMGA) effectively manage execution drag, yielding net strategy alphas that are statistically indistinguishable from zero ($\hat{\alpha} = -0.18\%, t = -0.03, p = 0.976$). The 5-factor model accounts for 41.2% of TFDMGA's return variance, with positive factor loadings on operating profitability (RMW beta $= +0.512, t = +9.12$).

Crucially, the regression specification in Table 4.6 evaluates the zero-investment Long-Short quintile spread portfolio ($Q_5 - Q_1$). Evaluating the Long-Short spread eliminates market beta exposure, yielding a moderate $R^2 = 41.2\%$ driven primarily by factor tilt (RMW and HML). By contrast, evaluating the long-only quintile portfolio (Q_5) against the market produces an $R^2 > 88.5\%$ dominated by $Mkt - RF$ systematic market exposure. This confirms that deep sequence architectures function as dynamic factor allocation engines rather than discoverers of unpriced market arbitrage (Fama and French, 2015; Kelly et al., 2019).

5. Conclusion & Future Extensions

5.1 Summary of Research Findings

In this thesis, I conducted an empirical investigation into whether statistical machine learning and deep learning models improve cross-sectional stock return prediction across S&P 500 constituent equities over the ten-year period from 2015 through 2024 (Gu et al., 2020; Chen et al., 2024). By constructing a point-in-time aligned panel free from lookahead and survivorship bias (McLean and Pontiff, 2016), evaluating models across 5 expanding-window walk-forward test folds (2020–2024) (López de Prado, 2018), accounting for 10 bps transaction cost drag (Novy-Marx and Velikov, 2016), and estimating Fama-French five-factor spanning regressions (Fama and French, 2015; Carhart, 1997), my study provides answers to all four core research questions:

1. **Statistical Performance of Deep Sequence Encoders (RQ1)**: Nonlinear sequence-based deep learning models outperform traditional linear regressions (Fama-MacBeth OLS, LASSO) and static tree classifiers (Random Forest, XGBoost). The proposed **TFDMGA network achieves high out-of-sample predictive accuracy**, yielding a daily Information Coefficient of $IC = +0.0348$, an annualized $ICIR = 3.12$, a ROC AUC of 0.6120, and a directional accuracy of 56.84%. A Diebold and Mariano (1995) test with Harvey et al. (1997) small-sample corrections indicates statistical significance over standard LSTM models ($DM = 2.41, p = 0.016$).

2. **Value of Multi-Modal Feature Fusion (RQ2)**: Combining multi-frequency feature modalities (16 Technical, 30 Fundamental, 2 Sentiment, 5 Macro = 53 ML features) through a directed ring attention cascade (Technical $\leftarrow$ Sentiment $\leftarrow$ Fundamental $\leftarrow$ Macro) (Vaswani et al., 2017) and dynamic macro gating (Lim et al., 2021) produces higher out-of-sample signal stability compared to single-modality models (Cong et al., 2021). Sentiment indicators and price momentum

drive high-frequency signal directionality, while quarterly accounting quality provides structural drawdown defense.

3. **Trading Performance & Risk Overlays (RQ3)**: Under 10 bps transaction costs and 1-day lagged execution (Novy-Marx and Velikov, 2016; Avramov et al., 2023), an initial **$1,000 USD deposit** (Jan 2020) grows to **$3,120.99 USD** under the baseline LSTM Q_5 long strategy. When paired with a 2:1 Take-Profit/Stop-Loss (TPSL) dynamic risk management overlay, drawdowns during macro stress periods (2020 COVID market decline, 2022 Fed rate hikes) are reduced, allowing capital to compound to **$4,368.50 USD** for LSTM and **$6,482.10 USD** for TFDMGA.

4. **Econometric Factor Spanning & Market Efficiency (RQ4)**: Fama-French 5-factor spanning regressions yield an estimated net strategy alpha of $\hat{\alpha} = -0.18\%$ per year ($p = 0.976, R^2 = 41.2\%$) (Fama and French, 2015; Newey and West, 1987). Because net alpha is statistically indistinguishable from zero, 100% of the strategy's return generation is accounted for by systematic factor exposures ($Mkt - RF, SMB, HML, RMW, CMA$). This suggests that machine learning models function as **dynamic factor allocation engines** (Kelly et al., 2019; Kozak et al., 2020) rather than discoverers of unpriced market arbitrage, maintaining consistency with market efficiency (Fama, 1970).

5.2 Theoretical and Practical Implications

The empirical conclusions of this study carry several implications for quantitative finance practitioners and academic researchers:

- **For Empirical Asset Pricing**: The collapse of LASSO under linear MSE loss highlights the importance of rank-based loss functions ($\mathcal{L}_{\text{rank}}, \mathcal{L}_{\text{IC}}$) when training machine learning architectures on noisy financial return targets (Tibshirani, 1996; Kozak et al., 2020; Gu et al., 2020).

- **For Institutional Asset Managers**: Daily alpha signals experience turnover drag ($\approx 13\%$ daily turnover). Incorporating dynamic risk management rules (such as 2:1

TPSL circuit breakers) is essential for preserving capital and converting statistical predictions into tradeable wealth compounding (Novy-Marx and Velikov, 2016; Frazzini and Pedersen, 2014).

- **For Market Efficiency Debates**: Demonstrating that deep learning model returns are spanned by systematic Fama-French factors provides insight into why ML strategies generate positive returns. Machine learning does not violate market efficiency (Fama, 1970); rather, it automates dynamic factor rotation across macroeconomic regimes (Kelly et al., 2019; Chen et al., 2024).

5.3 Limitations of the Study

While this thesis adheres to standard backtesting protocols (López de Prado, 2018), certain empirical limitations exist:

1. **Execution Slippage & Market Impact**: Transaction costs were modeled using fixed round-trip friction levels (5, 10, 20 bps). Real-world execution of large institutional orders introduces nonlinear square-root market impact drag (Novy-Marx and Velikov, 2016).

2. **Universe Limitation**: The empirical panel is restricted to U.S. Large-Cap S&P 500 equities. Smaller, less liquid mid-cap or small-cap stocks may present different return patterns along with higher transaction costs (Avramov et al., 2023; Hou et al., 2020).

5.4 Future Research Extensions

The empirical baseline established in this thesis provides several avenues for future research extensions:

Extension 1: Spatio-Temporal Graph Neural Networks – Extending the stock sequence encoders in TFDMGA to a Spatio-Temporal Graph Attention Network (GAT), modeling stocks as graph nodes with dynamic edge weights $A_{ij,t}$ to capture cross-sectional supply chain linkages, industry spillovers, and dynamic return correlations (Vaswani et al., 2017; Chen et al., 2024).

Extension 2: Macro-Conditioned Hyper-LoRA Attention – Conditioning Transformer attention projections (W_Q, W_K, W_V) on macroeconomic regimes using Hypernetworks and Low-Rank Adaptation (LoRA) to allow dynamic adjustment to changing macroeconomic environments (Lim et al., 2021).

Extension 3: Continuous Deep Reinforcement Learning with Differentiable QP Layers – Replacing decile portfolio sorting with a continuous Deep Reinforcement Learning (DRL) agent (PPO) coupled with a differentiable Quadratic Programming solver layer (`cvxpylayers`) to optimize continuous portfolio weights directly against nonlinear Sharpe ratio rewards (López de Prado, 2020; Novy-Marx and Velikov, 2016).

Extension 4: Conditional Variational Autoencoder Factor Extraction – Developing a probabilistic Conditional Variational Autoencoder (CVAE) to extract nonlinear latent factor spaces from high dimensional characteristics (Gu et al., 2021; Kelly et al., 2019; Chen et al., 2024).

Bibliography

Doron Avramov, Si Cheng, and Lior Metzker. Machine learning vs. economic restrictions: Evidence from stock return predictability. *Management Science*, 69(5):2587–2619, 2023. doi: 10.1287/mnsc.2022.4578.

Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint arXiv:1803.01271*, 1803(01271):1–14, 2018. doi: 10.48550/arXiv.1803.01271.

Fischer Black. Capital market equilibrium with restricted borrowing. *The Journal of Business*, 45(3):444–455, 1972. doi: 10.1086/295472.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324.

Mark M. Carhart. On persistence in mutual fund performance. *The Journal of Finance*, 52(1):57–82, 1997. doi: 10.1111/j.1540-6261.1997.tb03808.x.

Liying Chen, Markus Pelger, and Jason Zhu. Deep learning in asset pricing. *Management Science*, 70(2):714–750, 2024. doi: 10.1287/mnsc.2023.4716.

Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, volume 22, pages 785–794. ACM, 2016. doi: 10.1145/2939672.2939785.

Alex Chinco, Adam D. Clark-Joseph, and Mao Ye. Sparse signals in the cross-section of returns. *The Journal of Finance*, 74(1):449–492, 2019. doi: 10.1111/jofi.12733.

John H. Cochrane. Discount rates. *The Journal of Finance*, 66(4):1047–1108, 2011. doi: 10.1111/j.1540-6261.2011.01671.x.

Lin William Cong, Tengge Liang, and Xiao Zhang. Textual analysis in finance: A survey and new frontiers. *Journal of Financial Economics*, 142(2):512–538, 2021. doi: 10.1016/j.jfineco.2021.05.050.

Francis X. Diebold and Robert S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. doi: 10.1080/07350015.1995.10524599.

Eugene F. Fama. Efficient capital markets: A review of theory and empirical work. *The Journal of Finance*, 25(2):383–417, 1970. doi: 10.1111/j.1540-6261.1970.tb00518.x.

Eugene F. Fama and Kenneth R. French. The cross-section of expected stock returns. *The Journal of Finance*, 47(2):427–465, 1992. doi: 10.1111/j.1540-6261.1992.tb04398.x.

Eugene F. Fama and Kenneth R. French. Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56, 1993. doi: 10.1016/0304-405X(93)90023-5.

Eugene F. Fama and Kenneth R. French. A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22, 2015. doi: 10.1016/j.jfineco.2014.10.010.

Eugene F. Fama and James D. MacBeth. Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3):607–636, 1973. doi: 10.1086/260061.

Guanhao Feng, Stefano Giglio, and Dacheng Xiu. Taming the factor zoo: A test of new factors. *The Journal of Finance*, 75(3):1327–1370, 2020. doi: 10.1111/jofi.12883.

Andrea Frazzini and Lasse Heje Pedersen. Betting against beta. *Journal of Financial Economics*, 111(1):1–25, 2014. doi: 10.1016/j.jfineco.2013.10.005.

Joachim Freyberger, Andreas Neuhierl, and Michael Weber. Dissecting characteristics nonparametrically. *The Review of Financial Studies*, 33(5):2326–2377, 2020. doi: 10.1093/rfs/hhz123.

Jerome H. Friedman. Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232, 2001. doi: 10.1214/aos/1013203451.

Stefano Giglio and Dacheng Xiu. Asset pricing with omitted factors. *Journal of Political Economy*, 129(7):1947–2002, 2021. doi: 10.1086/714093.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2276, 2020. doi: 10.1093/rfs/hhaa009.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. Autoencoder asset pricing models. *Journal of Econometrics*, 222(1):429–450, 2021. doi: 10.1016/j.jeconom.2020.07.009.

Peter R. Hansen. A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380, 2005. doi: 10.1198/073500105000000063.

Campbell R. Harvey and Yan Liu. False discoveries in financial economics. *The Journal of Finance*, 75(5):2501–2553, 2020. doi: 10.1111/jofi.12888.

Campbell R. Harvey, Yan Liu, and Heqing Zhu. ...and the cross-section of expected returns. *The Review of Financial Studies*, 29(1):5–68, 2016. doi: 10.1093/rfs/hhv059.

David Harvey, Stephen Leybourne, and Paul Newbold. Testing the equality of prediction errors out of sample. *International Journal of Forecasting*, 13(2):281–291, 1997. doi: 10.1016/S0169-2070(96)00719-4.

Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (gelus). *arXiv preprint arXiv:1606.08415*, 2016.

Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.

Arthur E. Hoerl and Robert W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970. doi: 10.1080/00401706.1970.10488634.

Kewei Hou, Chen Xue, and Lu Zhang. Digesting anomalies: An investment approach. *The Review of Financial Studies*, 28(3):650–705, 2015. doi: 10.1093/rfs/hhu068.

Kewei Hou, Chen Xue, and Lu Zhang. Replicating anomalies. *The Review of Financial Studies*, 33(5):2019–2133, 2020. doi: 10.1093/rfs/hhz131.

Narasimhan Jegadeesh and Sheridan Titman. Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1):65–91, 1993. doi: 10.1111/j.1540-6261.1993.tb04702.x.

Dhiraj Kalamkar, Dheevatsa Mudigere, Naveen Mellempudi, Dipankar Kaul, Pradeep Banerjee, Sreenivas Avancha, Dharma Lu, Kaushik Vaidyanathan, Evangelos Georganas, Narayanan Sundaram, Anshumu Kapoor, Abhishek Rai, Dong-Ki Yang, Animesh Tiwari, Alexander Shiryaev, Alon Singer, and Mikhail Smelyanskiy. A study of bfloat16 for deep learning training. *arXiv preprint arXiv:1905.12322*, 2019.

Bryan Kelly, Seth Pruitt, and Yinan Su. Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):501–524, 2019. doi: 10.1016/j.jfineco.2019.05.001.

Serhiy Kozak, Stefan Nagel, and Shrihari Santosh. Shrinking the cross-section. *Journal of Financial Economics*, 135(2):271–292, 2020. doi: 10.1016/j.jfineco.2019.06.008.

Jonathan Lewellen, Stefan Nagel, and Jay Shanken. A skeptical guide to testing asset pricing models. *Journal of Financial Economics*, 96(2):175–194, 2010. doi: 10.1016/j.jfineco.2009.12.007.

Bryan Lim, Sören Ömer Arık, Nicolas Loeff, and Tomas Pfister. Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1393–1416, 2021. doi: 10.1016/j.ijforecast.2021.03.012.

John Lintner. The valuation of risk assets and the selection of risky investments in stock portfolios and capital budgets. *The Review of Economics and Statistics*, 47(1):13–37, 1965. doi: 10.2307/1924119.

Marcos López de Prado. *Advances in Financial Machine Learning*. John Wiley & Sons, Hoboken, NJ, 2018. doi: 10.1002/9781119482086.

Marcos López de Prado. *Machine Learning for Asset Managers*. Cambridge University Press, Cambridge, UK, 2020. doi: 10.1017/9781108883658.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *International Conference on Learning Representations (ICLR)*, 2019.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30, pages 4765–4774. Curran Associates, Inc., 2017. doi: 10.5555/3295222.3295230.

Harry Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952. doi: 10.1111/j.1540-6261.1952.tb01525.x.

R. David McLean and Jeffrey Pontiff. Does academic research destroy stock return predictability? *The Journal of Finance*, 71(1):5–32, 2016. doi: 10.1111/jofi.12365.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987. doi: 10.2307/1913605.

Robert Novy-Marx. The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1):1–28, 2013. doi: 10.1016/j.jfineco.2013.01.003.

Robert Novy-Marx and Mihail Velikov. A taxonomy of anomalies and their trading costs. *The Review of Financial Studies*, 29(1):104–147, 2016. doi: 10.1093/rfs/hhv063.

Ľuboš Pástor and Robert F. Stambaugh. Liquidity risk and expected stock returns. *Journal of Political Economy*, 111(3):642–685, 2003. doi: 10.1086/374184.

Stephen A. Ross. The arbitrage theory of capital asset pricing. *Journal of Economic Theory*, 13(3):341–360, 1976. doi: 10.1016/0022-0531(76)90046-6.

Jay Shanken. On the estimation of beta-pricing models. *The Review of Financial Studies*, 5(1):1–33, 1992. doi: 10.1093/rfs/5.1.1.

William F. Sharpe. Capital asset prices: A theory of market equilibrium under conditions of risk. *The Journal of Finance*, 19(3):425–442, 1964. doi: 10.1111/j.1540-6261.1964.tb02865.x.

Carolin Strobl, Anne-Laure Boulesteix, Achim Zeileis, and Torsten Hothorn. Bias in random forest variable importance measures: Illustrations, sources and a solution. *BMC Bioinformatics*, 8(1):25, 2007. doi: 10.1186/1471-2105-8-25.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1):267–288, 1996. doi: 10.1111/j.2517-6161.1996.tb02080.x.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008. Curran Associates, Inc., 2017. doi: 10.5555/3295222.3295349.

Halbert White. A reality check for data mining. *Econometrica*, 68(5):1097–1126, 2000. doi: 10.1111/1468-0262.00152.

Ruibin Xiong, Yining Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On layer normalization in the transformer architecture. *Proceedings of Machine Learning Research (PMLR)*, 119: 10524–10533, 2020.

Hui Zou and Trevor Hastie. Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2): 301–320, 2005. doi: 10.1111/j.1467-9868.2005.00503.x.